

A Brief History of Microbiology

The science of Microbiology dates back only two hundred years.

However, microorganisms have been around for thousands of years.

Ancestors of bacteria were the first living cells to appear on Earth.

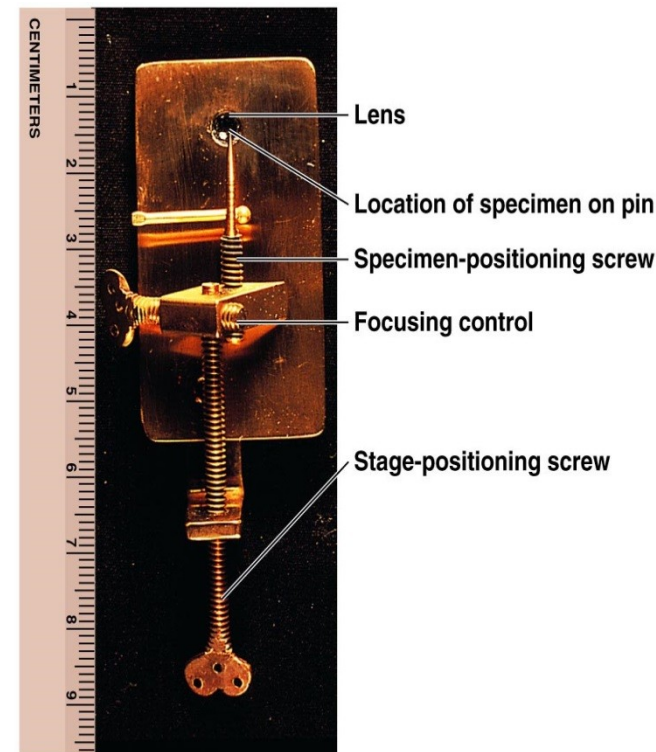
The first microbes (animalcules) were observed in 1673 by **Leeuwenhoek**.

In 1665, **Robert Hooke** reported that living things were composed of little boxes or cells, with the help of a relatively crude microscope.

In 1858, **Rudolf Virchow** said cells arise from preexisting cells.

Cell theory: All living things are composed of cells and come from preexisting cells.

1673-1723: **Antoni van Leeuwenhoek** described live microorganisms (animalcules) that he observed in teeth scrapings, rain water.



(b) Microscope replica

The Debate Over Spontaneous Generation

After van Leeuwenhoek discovered the “invisible” world of microorganisms, the scientific community of that time became interested in the origins of these tiny living things.

Not much more than 100 years ago, many scientists and philosophers believed that some forms of life could arise spontaneously from nonliving matter, they called this the hypothesis of **spontaneous generation**.

Therefore, people commonly believed that toads, snakes, and mice could be born of moist soil; that flies could emerge from manure; and that maggots, the larvae of flies, could arise from decaying matters.

According to spontaneous generation, a “**vital force**” forms life.

The alternative hypothesis, that the living organisms arise from preexisting life, is called **biogenesis**.

Evidence PRO and CON to spontaneous generation

Redi's Experiments

In 1668: A strong opponent of SG, **Francisco Redi** set out to demonstrate that maggots did not arise spontaneously from decaying meat.

Redi filled two jars with decaying meat.

The first was left unsealed; the flies laid their eggs on the meat, and the eggs developed into larvae.

The second jar was sealed and, because the flies couldn't lay their eggs on the meat, no maggots appeared.

- Redi's antagonists were not convinced; they claimed that fresh air was needed for spontaneous generation.
- **Redi set up a second experiment, in which**
a jar was covered with a fine net instead of being sealed.

No larvae appeared in the gauze-covered jar, even though air was present.

Maggots appeared only when flies were allowed to leave eggs on the meat.

Redi's results blew the belief that large forms of life could arise from nonlife.

The Theory of Biogenesis

Pasteur next placed broth in open-ended long-necked flasks and bent the necks into S-shaped curves.

The contents of these flasks were then boiled and cooled.

The broth of in the flasks did not decay and showed no signs of life.

Pasteur's S-shaped neck allowed air to pass into the flask, but trapped the airborne MOs that might contaminate the broth.

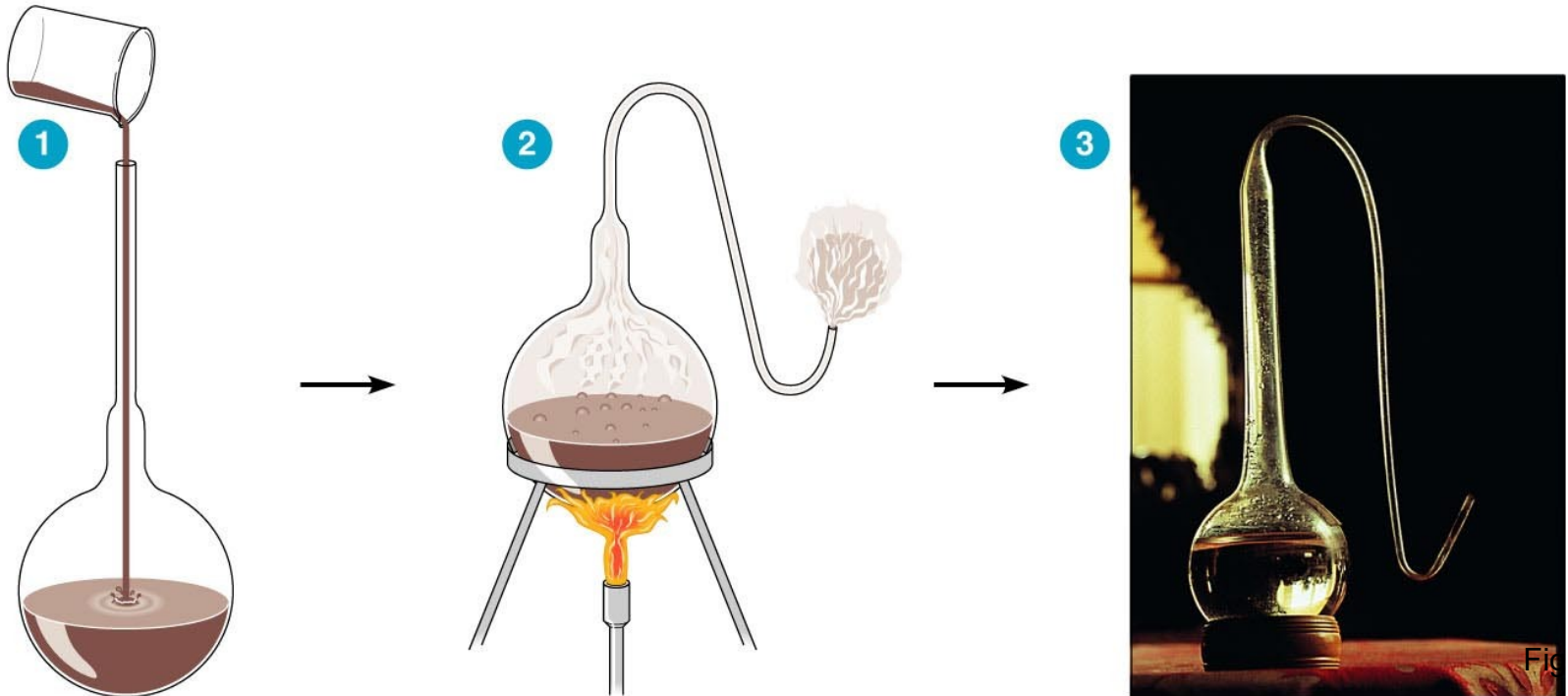


Figure 1.3

Pasteur's Findings

Pasteur showed that MOs can be present in nonliving matter- on solids, in liquids, and in the air.

He demonstrated that microbial life can be destroyed by heat and devised methods to block access of airborne MOs to nutrients.

These discoveries forms the basis of **aseptic techniques** (techniques that prevent contamination by unwanted MOs.), which are now the standard practice in laboratory and many medical procedures.

Pasteur's work provided evidence that MOs can not originate from mystical forces preset in nonliving materials.

Scientists now believe that a form of spontaneous generation probably did occur on primitive Earth when life first began.

Pasteur showed that microbes are responsible for fermentation.

The Golden Age of Microbiology

The period from 1857-1914, has been named the **Golden Age of Microbiology**.

During this period, rapid advances headed by **Pasteur** and **Robert Koch**, led to the establishment of microbiology as a science.

Beginning with Pasteur's work, discoveries included

- The agents of many diseases.

- The role of immunity in the prevention and cure of diseases.

- The relationship between microbes and disease.

- Antimicrobial drugs

- Improved the techniques for microscopy and culturing microorganisms.

- Development of vaccines and surgical techniques.

- Studying the chemical activities of microorganisms.

Louis Pasteur (1822 - 1895)

trapped airborne organisms in cotton;

he also heated the necks of flasks, drawing them out into long curves, sterilized the media, and left the flasks open to the air;

no growth was observed because dust particles carrying organisms did not reach the medium, instead they were trapped in the neck of the flask; if the necks were broken, dust would settle and the organisms would grow; in this way Pasteur disproved the theory of spontaneous generation

Joseph Lister (1827 - 1912)

Developed a system of surgery designed to prevent microorganisms from entering wounds – phenol sprayed in air around surgical incision

Decreased number of post-operative infections in patients

His published findings (1867) transformed the practice of surgery

Robert Koch (1843 - 1910),

using criteria developed by his teacher, Jacob Henle (1809-1895), established the relationship between *Bacillus anthracis* and anthrax;

his criteria became known as Koch's

Postulates and are still used to establish the link between a particular microorganism and a particular disease:

Koch's Postulates

The causative (etiological) agent must be present in all affected organisms but absent in healthy individuals

The agent must be capable of being isolated and cultured in pure form

When the cultured agent is introduced to a healthy organism, the same disease must occur

The same causative agent must be isolated again from the affected host

Development of Culture Media

Why?

To enable the isolation of pure cultures (only one type of organism)

Especially important during Koch's period

Gelatin not useful as solidifying agent (melts at >28 degrees Celsius and some bacteria hydrolyze it with enzymes)

Fannie Hesse, the wife of one of Koch's assistants, proposed using agar

- Not digested by most bacteria

- Melts at 100 degrees Celsius

- Used today - ~2% in solid media

Richard Petri, another of Koch's assistants, developed the Petri dish

Development of Vaccines and Antisera

Edward Jenner in 1796 discovered that cowpox (vaccinia) induced protection against human smallpox

Called procedure vaccination

Vaccination:

Inoculation of healthy individuals with weakened (or attenuated) forms of microorganisms, that would otherwise cause disease, to provide protection, or active immunity from disease upon later exposure

Pasteur and Roux reported that incubating cultures longer than normal in the lab resulted in ATTENUATED bacteria that could no longer cause disease

Working with chicken cholera (caused by *Pasteurella multocida*), they noticed that animals injected with attenuated cultures were resistant to the disease

Louis Pasteur

demonstrated that alcoholic fermentations were the result of microbial activity,

that some organisms could decrease alcohol yield and sour the product, and

that some fermentations were aerobic and some anaerobic;

he also developed the process of pasteurization to preserve wine during storage

Fermentation and Pasteurization

At that time, many scientists believed that air converted the sugars in beverages into alcohols.

Pasteur found instead that microbes called yeasts **convert the sugars to alcohols in the absence of air in a process called fermentation.**

Fermentation is the conversion of sugar to alcohol to make beer and wine.

Souring and spoilage are caused by different MOs called bacteria.

Showing the connection between spoilage of food and MOs was a major step towards establishing the relationship between disease and microbes.

In the presence of air, bacteria change the alcohol in the beverage into vinegar (acetic acid).

Pasteur's solution to the spoilage problem was to heat the beer and wine just enough to kill most of the bacteria that caused the spoilage in a process called **pasteurization**.

Pasteurization is now commonly used to reduce spoilage and kill potentially harmful bacteria in milk as well as in some alcoholic drinks.

The Germ Theory of Disease

Until relatively recently, the fact that many kinds of diseases are related to MOs was unknown.

Before the time of Pasteur, effective treatments for many diseases were discovered by **trial and error**, but the causes of the diseases were unknown.

The realization that yeasts play a crucial role in fermentation was the first link between the activity of a MO and physical and chemical changes in organic materials.

This discovery alerted scientists that MOs might have similar relationships with plants and animals- specially, that MOs might cause diseases.

This idea was known as the germ theory of disease.

Many people did not accept this theory at that time, because for centuries disease was believed to be punishment for individual's crimes and misdeeds.

Most people in Pasteur's time found it inconceivable that "invisible" microbes could travel through the air to infect plants and animals, or remain on clothing and bedding to be transmitted from one person to another.

The Germ Theory of Disease

1860s: **Joseph Lister** used a chemical disinfectant (phenol) to prevent surgical wound infections after looking at Pasteur's work showing microbes are in the air, can spoil food, and cause animal diseases.

1876: **Robert Koch** proved for the first time that a bacterium causes anthrax and provided the experimental steps, Koch's postulates, to prove that a specific microbe causes a specific disease.

Vaccination

1796: **Edward Jenner** found a way to protect people from smallpox almost 70 years before Koch established that microorganism causes anthrax.

He inoculated a healthy 8-years-old volunteer with cowpox virus. The person was then protected from cowpox and smallpox.

The process was called Vaccination, derived from Latine word vacca for cow.

The protection from disease provided by vaccination or by recovery from the disease itself is called immunity.

In about 1880, Pasteur discovered why vaccination work by working on cholera vaccination.

Pasteur used the term vaccine for cultures of avirulent microorganisms used for preventive inoculation.

Some vaccines are still produced from avirulent microbial strains, others are made from killed virulent microbes, from isolated components of virulent MOs, or by genetic engineering techniques.

The Birth of Modern Chemotherapy

Treatment of disease by using chemical substances is called chemotherapy.

Chemotherapeutic agents prepared from chemicals in the laboratory are called **synthetic drugs**.

Chemotherapeutic agents produced naturally by bacteria and fungi to act against other MOs are called **antibiotics**.

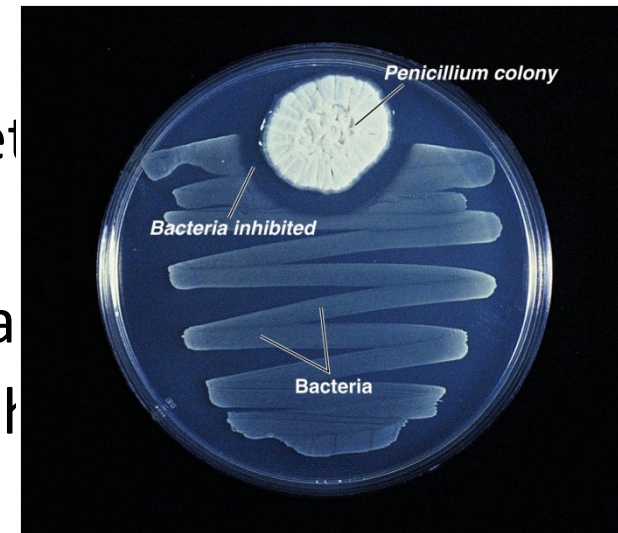
The success of chemotherapy is based on the fact that some chemicals are more poisonous to MOs than to the hosts infected by the microbes.

Quinine from tree bark was long used to treat malaria.

1910: **Paul Ehrlich** developed the first synthetic drug, **Salvarsan**, to treat syphilis. (the magic bullet!) **1930s**: Several other synthetic drugs derived from dyes that could destroy MOs were developed. **Sulfonamides** (sulfa drugs) were synthesized at about the same time.

The Birth of Modern Chemotherapy

- 1928: **Alexander Fleming** discovered the first antibiotic.
 - On a contaminated plate, around the mold (*Penicillium*) was a clear area where bacterial growth had been inhibited.
 - He observed that the *Penicillium* mold made an antibiotic, penicillin, that killed *Staphylococcus aureus*.
- 1940s: Penicillin was tested clinically and mass produced.
- Since then, thousands of antibiotics have been discovered.
- Antibiotics and other chemotherapeutic drug faces many problem:
 - Toxicity to humans in practical use, specially antiviral drugs
 - The emergence and spread of new varieties of MOs that are resistant to antibiotics.
(due to bacterial enzymes that inactivate a or prevention of antibiotic from entering the cell)



Modern Developments in Microbiology

Branches of Microbiology

- **Virology** is the study of viruses.

In 1892, **Dimitri Iwanowski** reported that the organism that caused mosaic disease of tobacco was so small that it passed the bacterial filters.

In 1935, **Wendell Stanley** demonstrated that the organism, called tobacco mosaic virus (TMV), was different from other microbes, so simple, and composed of only nucleic acid core and protein core.

In 1940s, the development of electron microscope enabled the scientists to observe the structure and activity of viruses in detail.

Sergei Winogradsky (1856 - 1953)

worked with soil bacteria and discovered that they could oxidize iron, sulfur, and ammonia to obtain energy;
he also studied anaerobic nitrogen-fixation and cellulose decomposition

Martinus Beijerinck (1851 - 1931)

Isolated aerobic nitrogen-fixing soil bacteria (*Azotobacter* and *Rhizobium*) and sulfate reducing bacteria

Beijerinck and Winogradsky

Pioneered the use of enrichment cultures and selective media

Modern Developments in Microbiology

Branches of Microbiology

- **Recombinant DNA Technology:**
 - In the 1960s, **Paul Berg** inserted animal DNA into bacterial DNA and the bacteria produced an animal protein.
 - Recombinant DNA is DNA made from two different sources.
 - Recombinant DNA technology, or genetic engineering, involves microbial genetics and molecular biology.
- Using microbes
 - Beadle and Tatum** showed that genes encode a cell's enzymes (1942).
 - Avery, MacLeod, and McCarty** showed that DNA was the hereditary material (1944).
 - Lederberg and Tatum** discovered that genetic material could be transferred from one bacterium to another by conjugation (1946).
 - Watson and Crick** proposed a model for the structure of DNA (1953).
 - Jacob and Monod** discovered the role of mRNA in protein synthesis (1961).

Size of the microorganisms

Microorganisms range in size from the smallest [virus at $0.01\mu\text{m}$ or 10 nm] to the largest [protozoa at $100\mu\text{m}$].

In order to view most of these microorganisms, an instrument called the microscope is used. Specific microscopy techniques involve light microscopy, scanning electron microscopy (SEM), and transmission electron microscopy (TEM).

MICROSCOPY

Microscopes are essential for microbiological studies

Light microscopes: cellular resolution

- bright-field (stains)

- dark-field

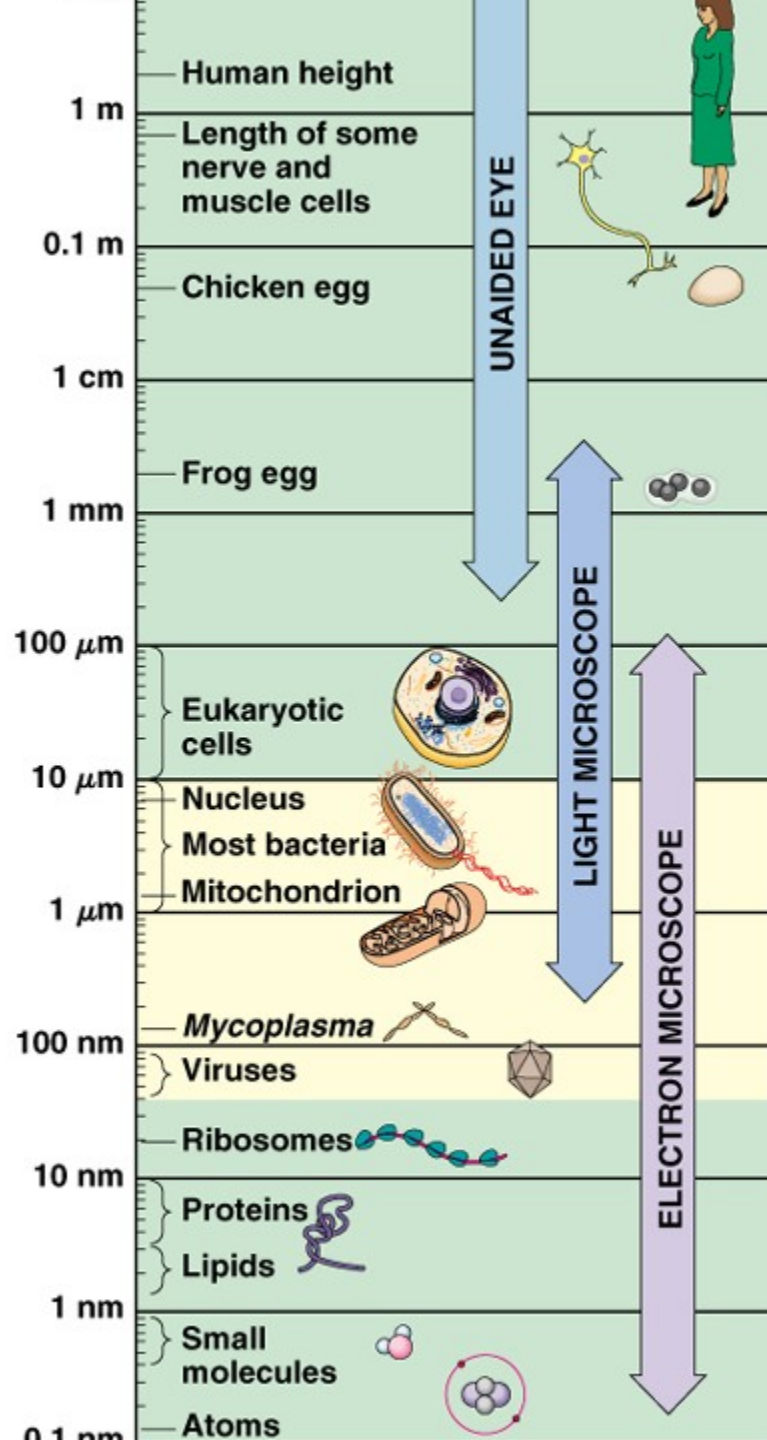
- phase contrast

- fluorescence (stains)

are compound microscopes

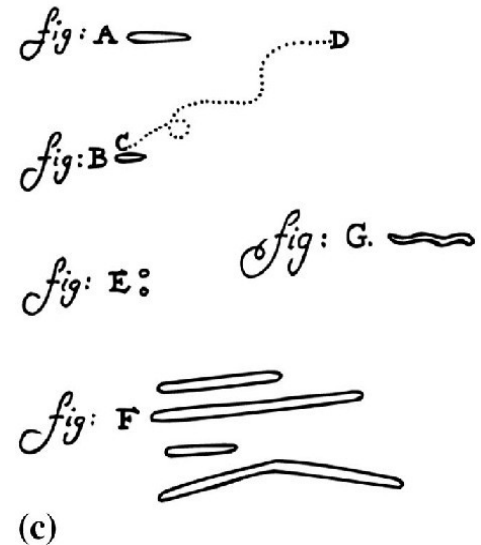
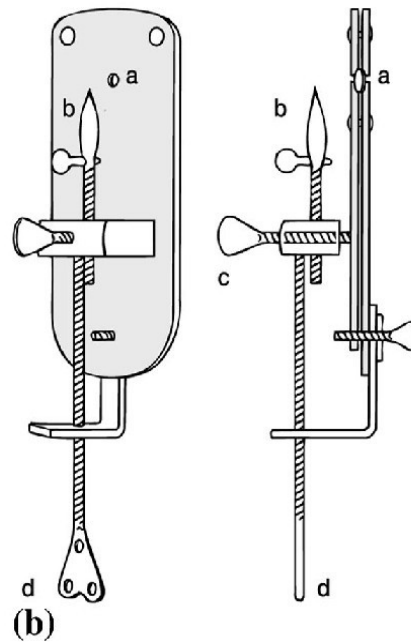
- image formed by action of ≥ 2 lenses

Electron microscopes: subcellular resolution



Discovery of Microorganisms

- Antony van Leeuwenhoek (1632-1723)
 - First person to observe and describe microorganisms accurately



Lenses and the Bending of Light

Light is refracted (bent) when passing from one medium to another

Refractive index

a measure of how greatly a substance slows the velocity of light

Direction and magnitude of bending is determined by the refractive indices of the two media forming the interface

Refractive Index of air - 1.00

water- 1.33

glass-1.5

immersion oil- 1.5

Lenses

Focus light rays at a specific place called the **focal point**

distance between center of lens and focal point is the **focal length**

Strength of lens related to focal length

short focal length $\Rightarrow$ more magnification

The Bright-Field Microscope

Produces a dark image against a brighter background

Has several objective lenses

parfocal microscopes remain in focus when objectives are changed

Total magnification

product of the magnifications of the ocular lens and the objective lens

Light Microscopy: Optics

Visualization depends on magnification (lenses) and resolution (physical properties of light)

The limit of resolution for a light microscope is about $0.2\text{ }\mu\text{m}$ (or 200 nm).

Objects closer than $0.2\text{ }\mu\text{m}$ cannot be resolved

Total magnification is product of the magnification of its ocular and its objective lenses.

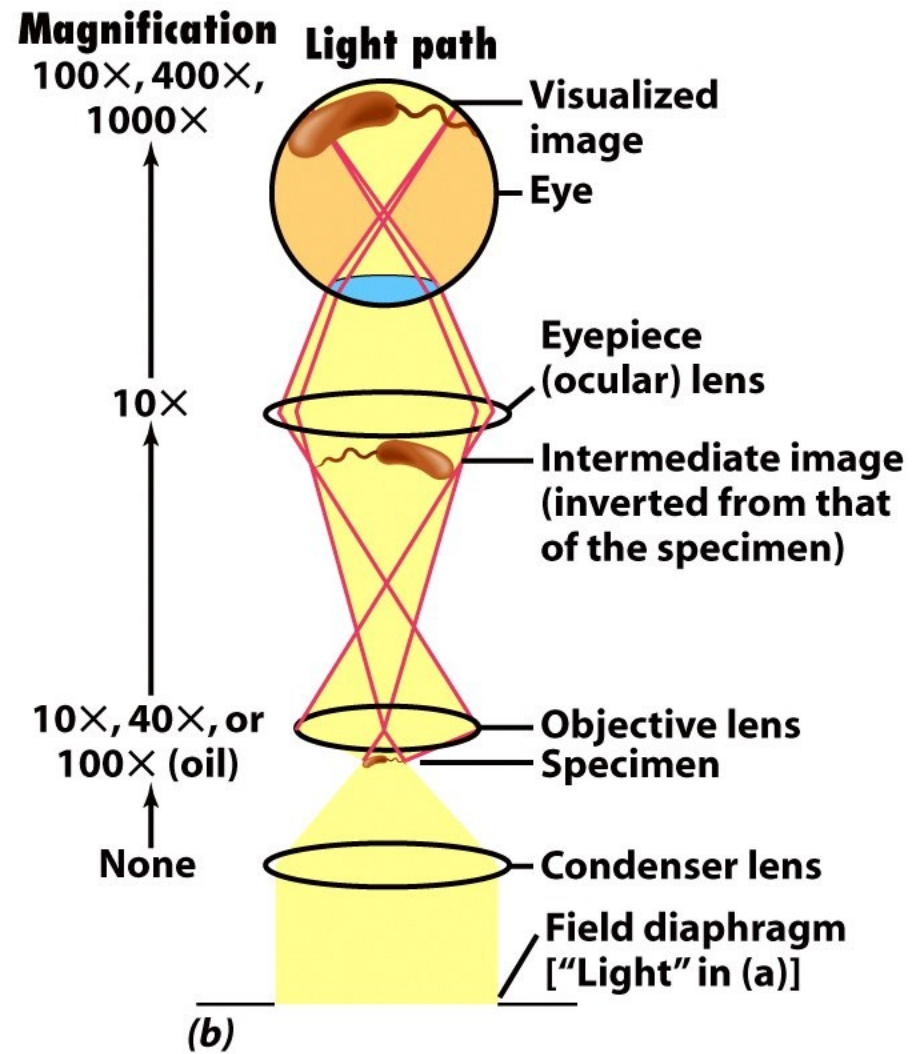
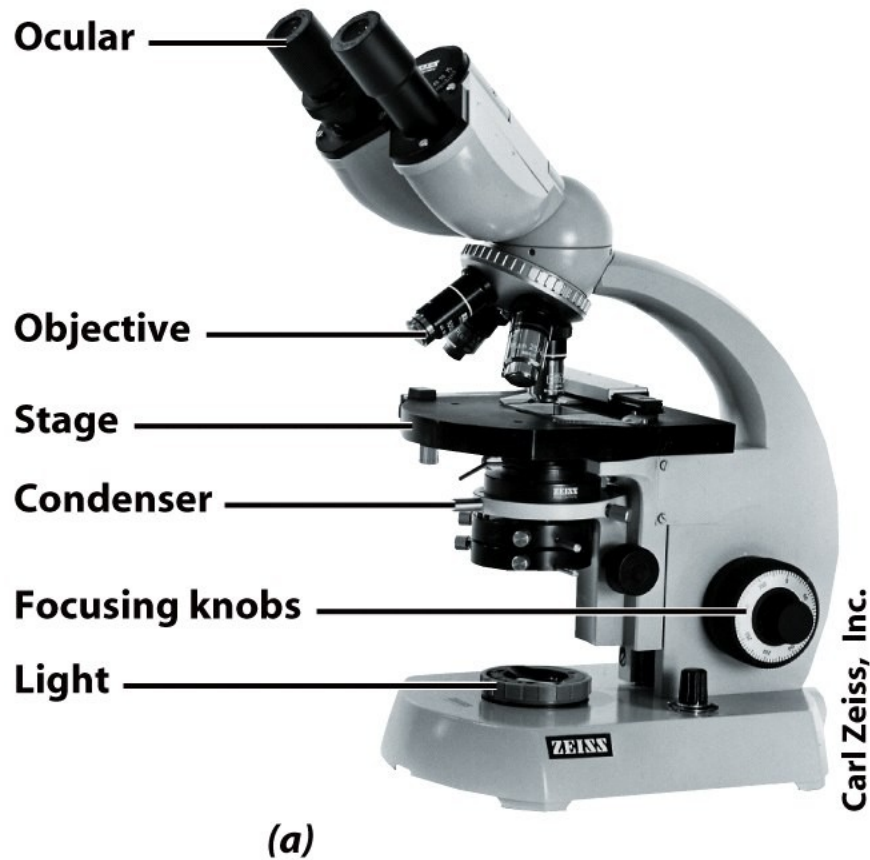
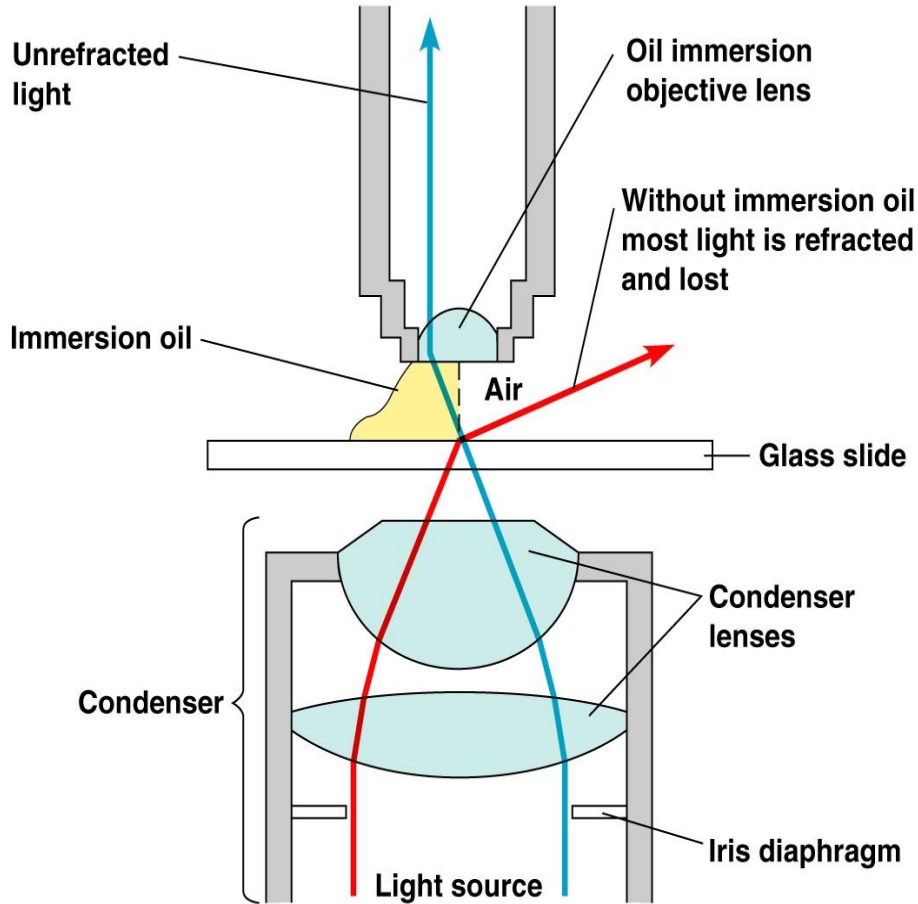


Figure 4-1 Brock Biology of Microorganisms 11/e
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Refractive Index



Measures light-bending ability of a medium

Light may bend in air so much that it misses the small high-magnification lens.

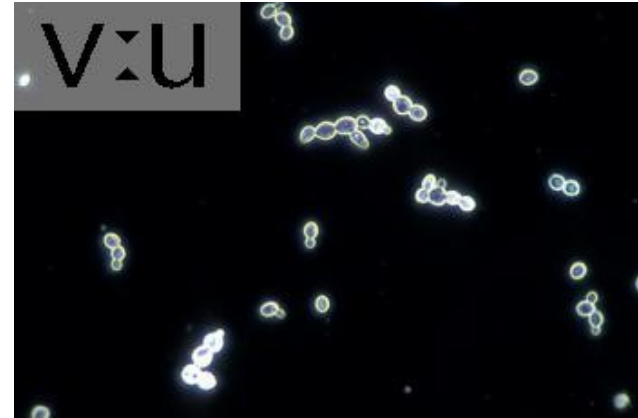
Immersion oil is used to keep light from bending.

Microscopy: Dark Field

Greater resolution

Light reaches specimens only from the side

Only the specimen itself is illuminated



Candida sp.

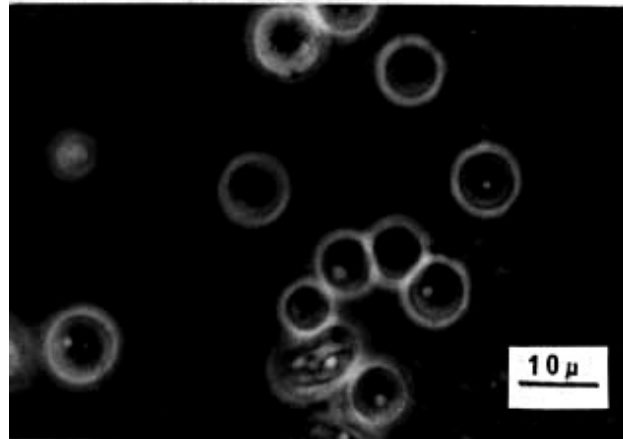
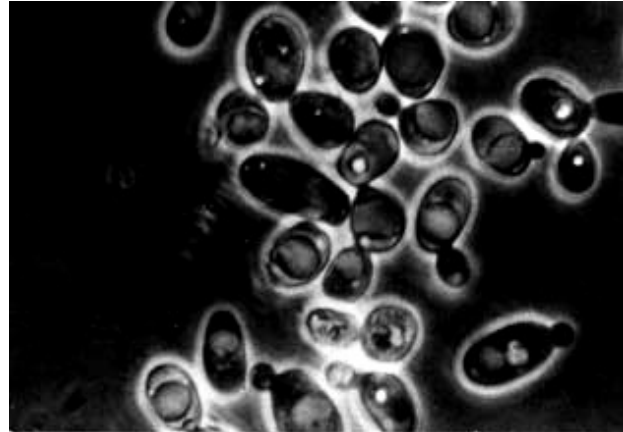


Treponema pallidum

Microscopy: Phase Contrast

May be used to visualize live samples and avoid distortion from cell stain

Image contrast is derived from the differential refractive index of cell structures.

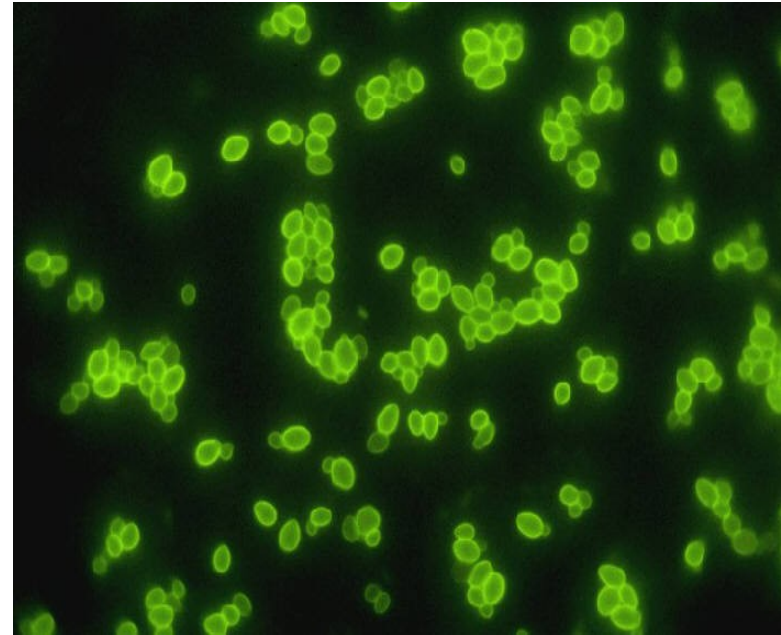


Microscopy: Fluorescence

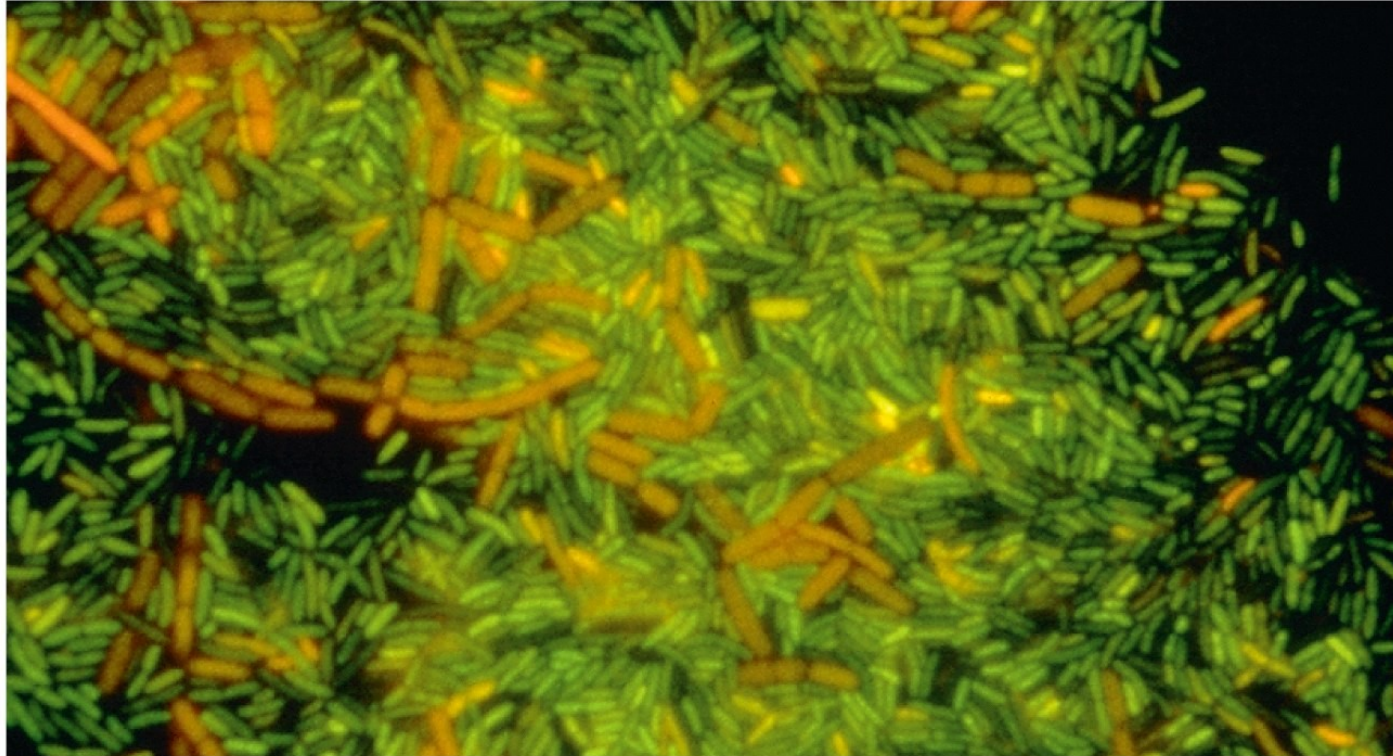
Visualization of autofluorescent cell structures (e.g., chlorophyll) or fluorescent stains

Can greatly increase the resolution of cells and cell structures

Many functional probes available



Example for Differential Fluorescence Stain



Molecular Probes, Inc., Eugene, Oregon

Figure 4-4c Brock Biology of Microorganisms 11/e
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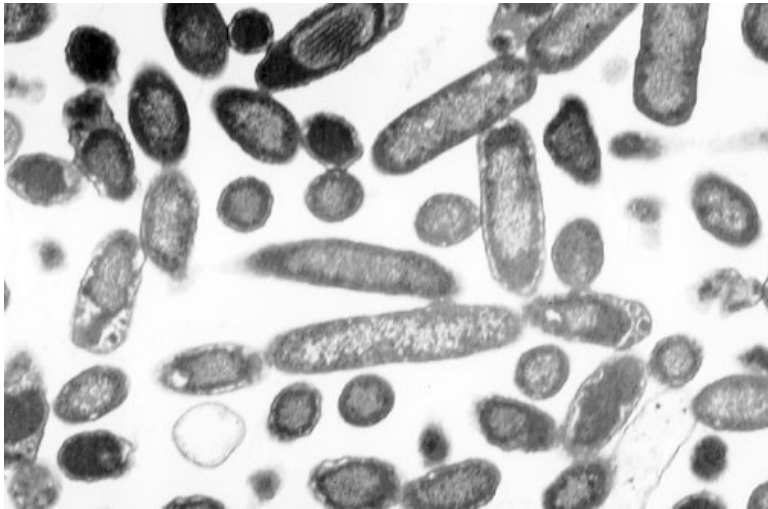
Pseudomonas (green) *Bacillus* (orange)

Microscopy: Electron Microscopy

- Electron microscopes have far greater resolving power than light microscopes, with limits of resolution of about 0.2 nm
- Two major types of electron microscopes
 - Transmission electron microscopy (TEM) for observing internal cell structure down to the molecular level
 - Scanning electron microscopy (SEM) for 3-D imaging and examining surfaces

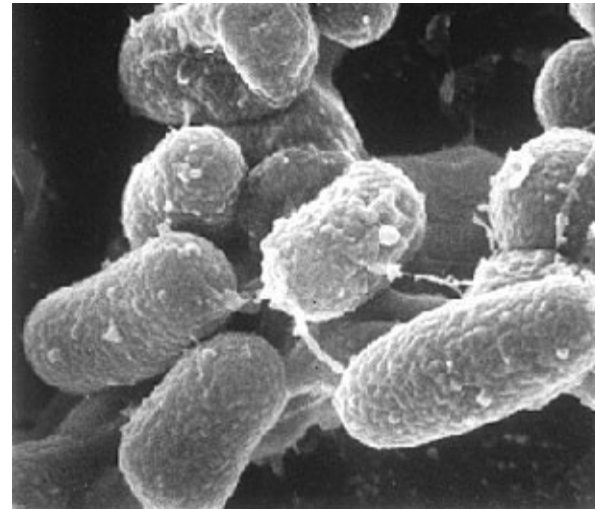
Electron Microscopy

TEM



Pseudomonas

SEM

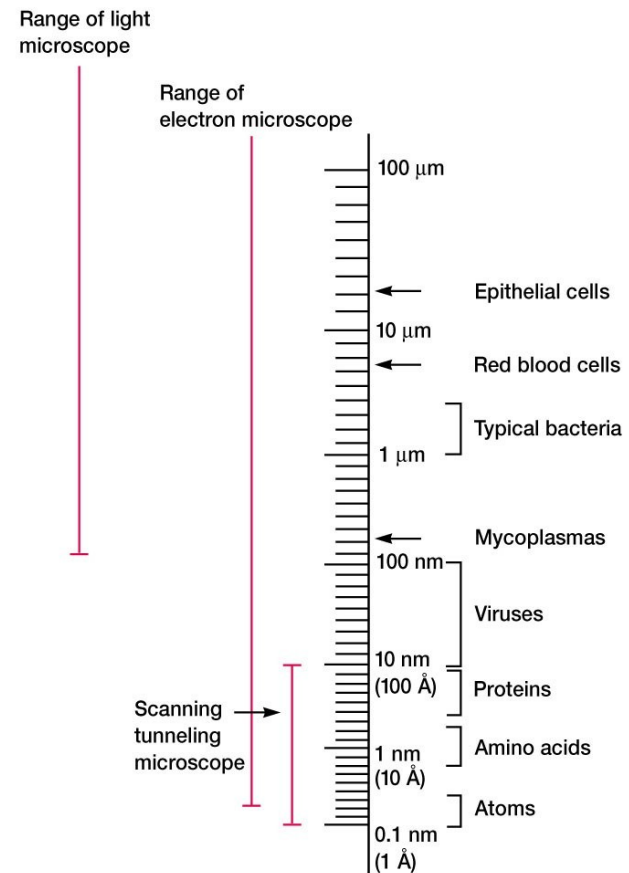


Mycobacterium

Electron Microscopy

Beams of electrons are used to produce images

Wavelength of electron beam is much shorter than light, resulting in much higher resolution

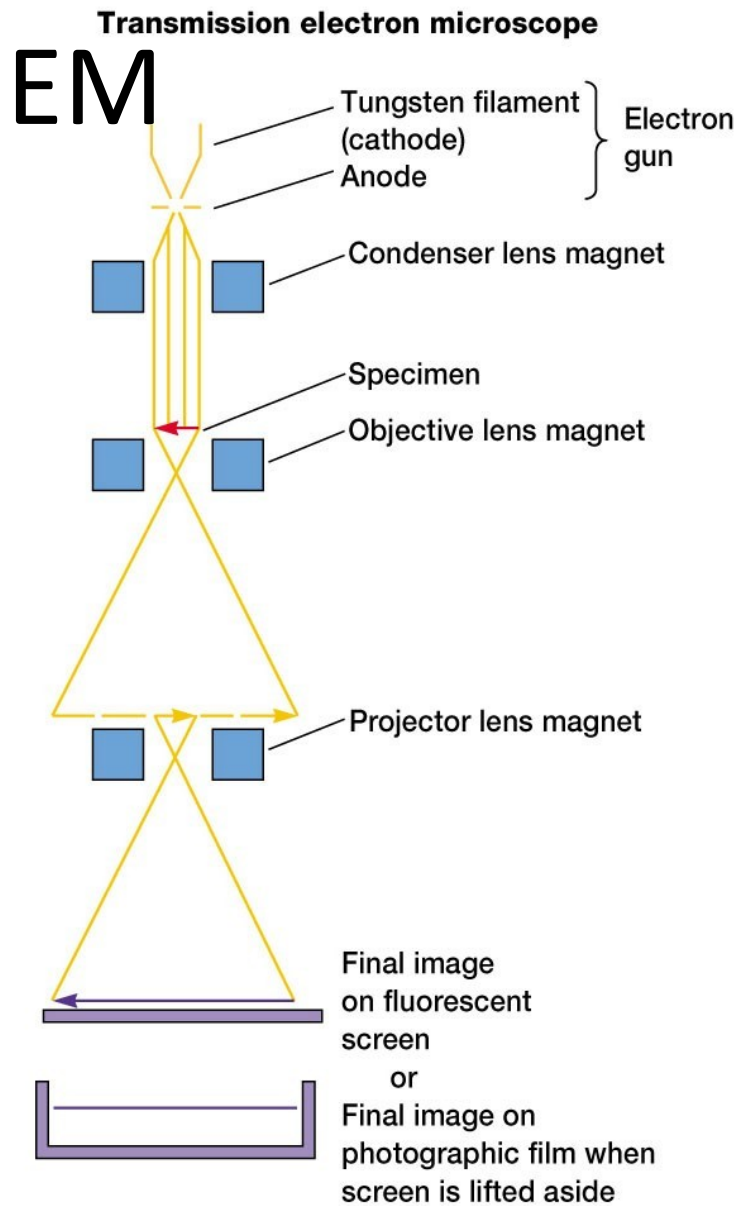


The Transmission Electron Microscope

Electrons scatter when they pass through thin sections of a specimen

Transmitted electrons (those that do not scatter) are used to produce image

Denser regions in specimen, scatter more electrons and appear darker



Specimen Preparation

Analogous to procedures used for light microscopy
for transmission electron microscopy, specimens must be cut very thin.

Specimens are chemically fixed and stained with electron dense material

Other preparation methods

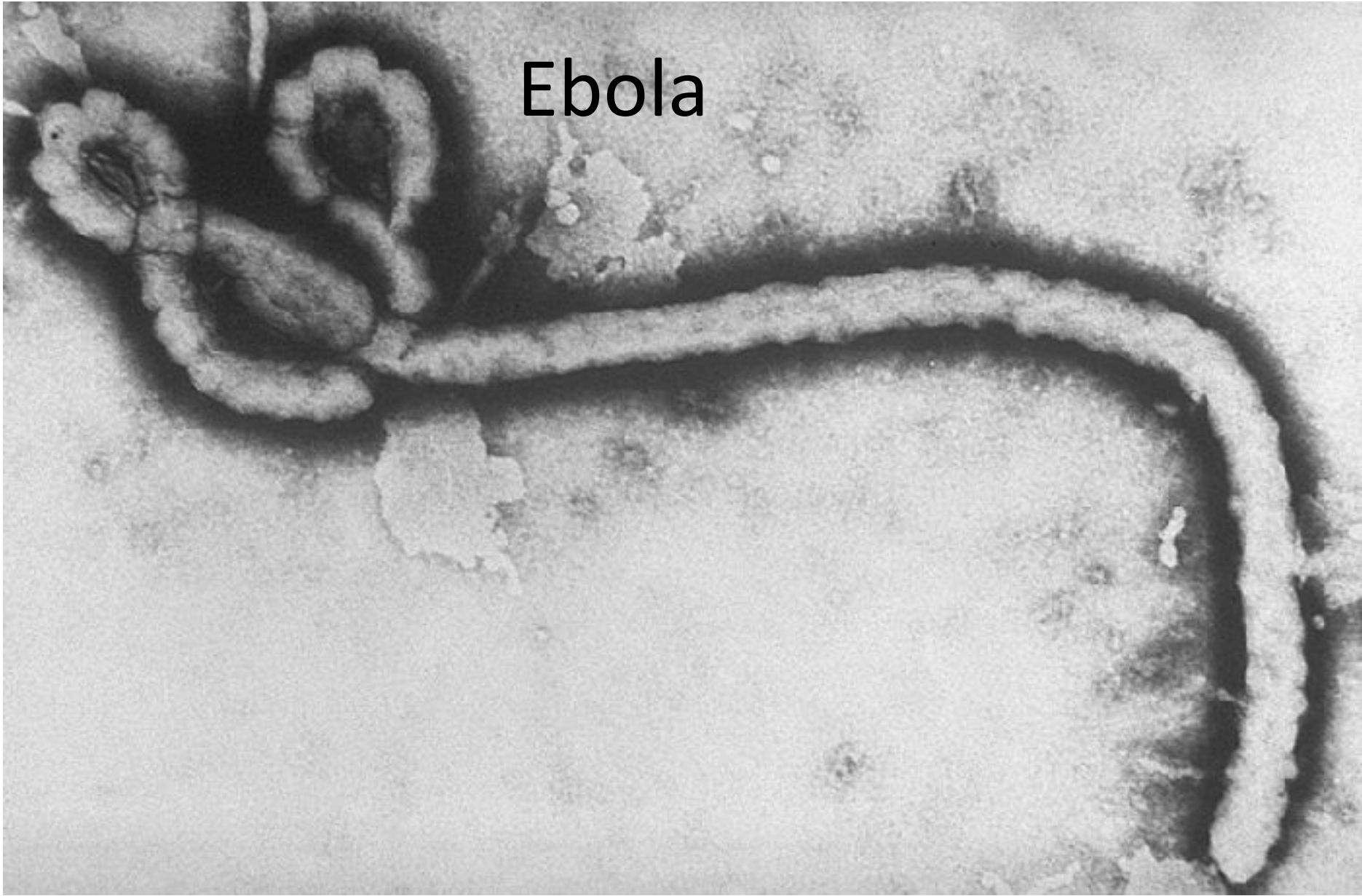
shadowing

coating specimen with a thin film of a heavy metal

freeze-etching

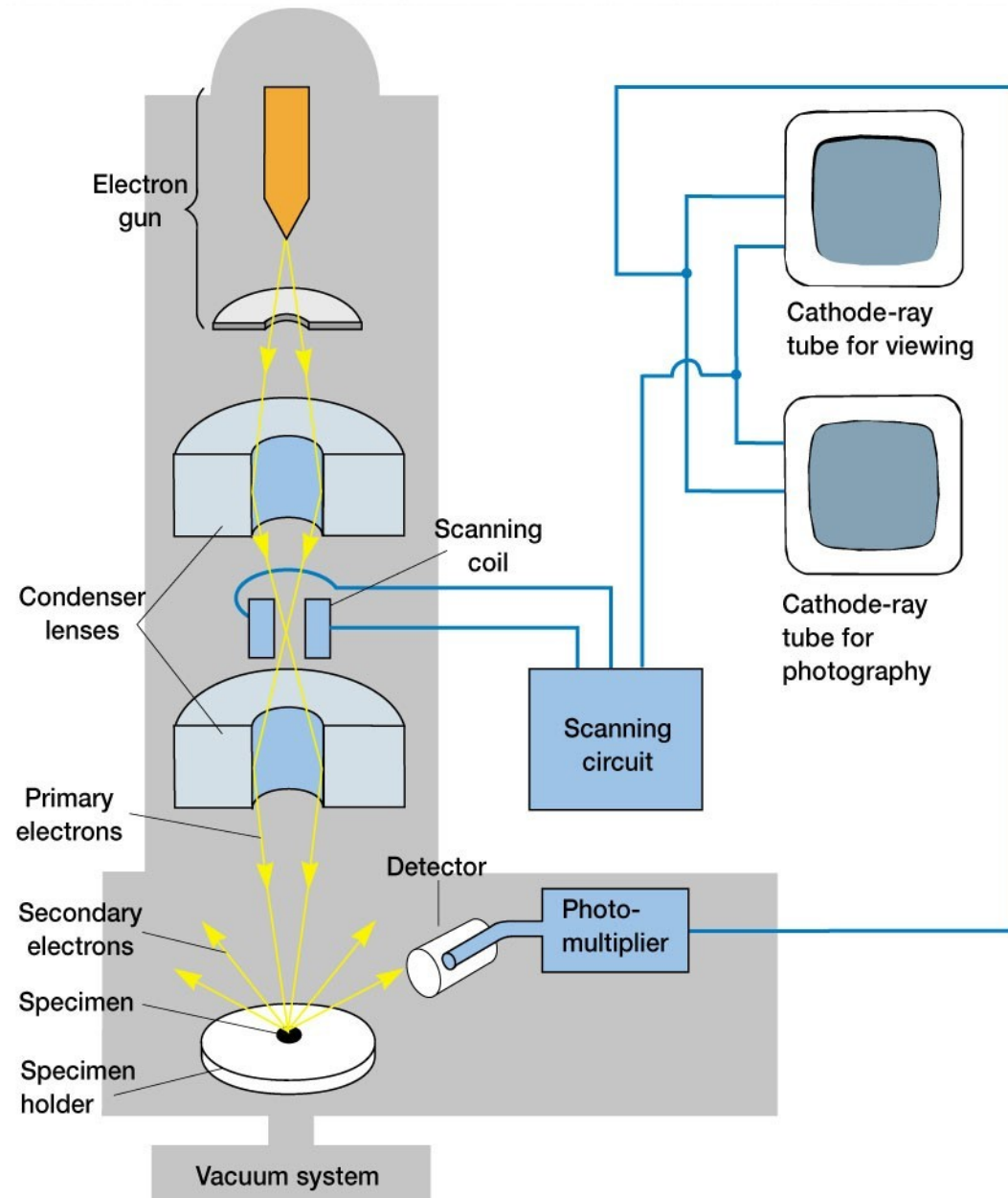
freeze specimen then fracture along lines of
greatest weakness (e.g., membranes)

Ebola



The Scanning Electron Microscope

- uses electrons reflected from the surface of a specimen to create image
- produces a 3-dimensional image of specimen's surface features

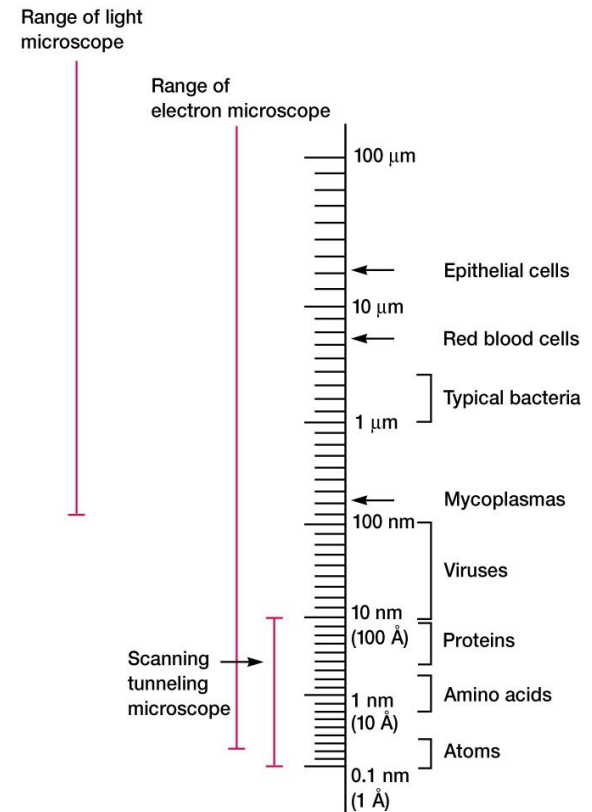


Newer Techniques in Microscopy

confocal microscopy
and scanning probe
microscopy

have extremely high
resolution

can be used to observe
individual atoms



Confocal Microscopy

confocal scanning laser microscope

laser beam used to illuminate spots on specimen

computer compiles images created from each point to generate a 3-dimensional image

Scanning Probe Microscopy

scanning tunneling microscope

- steady current (tunneling current) maintained between microscope probe and specimen
- up and down movement of probe as it maintains current is detected and used to create image of surface of specimen

atomic force microscope

- sharp probe moves over surface of specimen at constant distance
- up and down movement of probe as it maintains constant distance is detected and used to create image

PREPARING SPECIMEN FOR OBSERVATION UNDER A LIGHT COMPOUND MICROSCOPE

There are two ways to prepare a specimen to be observed under a light compound microscope. These are a smear and a wet mount.

Smear

A smear is a preparations process where a specimen that is spread on a slide. You prepare a smear using the heat fixation process.

1. Use a clean glass slide.
2. Take a smear of the culture. (The microorganisms are spread over the glass slide).
3. Place the live microorganism on the glass slide by smearing it onto the glass
4. The slice is air dried then passed over a Bunsen burner about three times.
5. The heat causes the microorganisms to adhere to the glass slide. This is known as fixing the microorganisms to the glass slide.
6. Stain the microorganisms with an appropriate stain.

Wet Mount

A wet mount is a preparation process where a live specimen in culture fluid is placed on a concave glass slide or a plain glass slide.

The concave portion of the glass tube forms a cup-like shape that is filled with a thick substance, such as carboxymethyl cellulose. The microorganism is free to move about within the fluid, although the viscosity of the substance slows its movement.

This makes it easier for you to observe the microorganism. The specimen and the substance are protected from spillage and outside contaminants by a glass cover that is placed over the concave portion of the slide.

Staining procedures

Staining is used to increase the contrast and resolution of a specimen.

Definition: Coloring the microorganism with a dye that emphasizes certain structures.

Preparation for Staining:

- 1) Create smear: film of specimen spread over slide and allowed to air dry
- 2) Fixing: adhere organism to slide by heat. Also kills microorganism
- 3) Stain: dyes applied and washed off.

Preparation and Staining of Specimens

Increases visibility of specimen

Denotes specific morphological features

Preserves specimens

Fixation

process by which internal and external structures are preserved and fixed in position

process by which organism is killed and firmly attached to microscope slide

heat fixing

preserves overall morphology but not internal structures

chemical fixing

protects fine cellular substructure and morphology of larger, more delicate organisms

Dyes and Simple Staining

dyes

make internal and external structures of cell
more visible by increasing contrast with
background

have two common features

chromophore groups

- chemical groups with conjugated double bonds
- give dye its color
- ability to bind cells

Types of Dyes

Stains are salts composed of positive and negative ions.
One of these ions is colored and called the **chromophore**.

Basic dyes consist of the positive ion that stains negatively charged structures.

Acid dyes consist of the negative ion that stains positively charged structures

Examples

Basic Dyes:

- Crystal violet (violet)
- Methylene blue (blue)
- Malachite Green (green)
- Safranin (red)

Acidic Dyes:

- Nigrosin (black)
- Congo Red (red)
- Eosin (red)
- Carbol (Acid) Fuschin (red)

Since bacteria are negative charged, **basic dyes** are used since the positive ion in the dye is attracted to the bacteria.

Acid dyes are not normally used since the dye's negative ion is repelled by the bacteria's negatively charged cytoplasm.

Acid dyes are used in a **negative staining** technique, in which the bacterial cells remain clear against a colored background.

Since the bacteria are not heat fixed, the cells normal size and shape are maintained.

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STAINING A SPECIMEN

Not all specimens can be clearly seen under a microscope. Sometimes the specimen blends with other objects in the background because they absorb and reflect approximately the same light waves. You can enhance the appearance of a specimen by using a stain.

A stain is used to contrast the specimen from the background.

A stain is a chemical that adheres to structures of the microorganisms and in effect dyes the microorganisms so it can be easily seen under a microscope. Stains used in microbiology are either basic or acidic.

Basic stains are cationic and have positive charge. Common basic stains are methylene blue, crystal violet, safranin, and malachite green. These are ideal for staining chromosomes and the cell membranes of many bacteria.

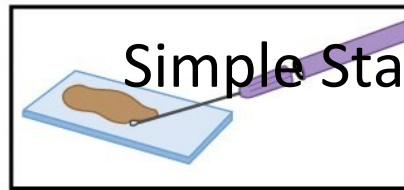
Acid stains are used to identify bacteria that have a waxy material in their cell walls. This form of staining differentiates bacteria.

Types of Stains

There are two types of Stains: simple and differential.

Simple Stain

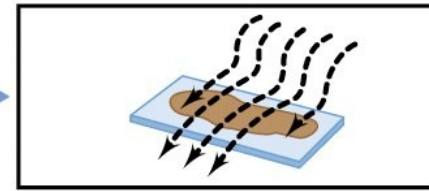
A simple stain has a simple basic dye that is used to show shapes of cells and the structures within a cell. Methylene blue, safranin, carbolfuchsin and crystal violet are common simple stains that are found in most microbiology laboratories.



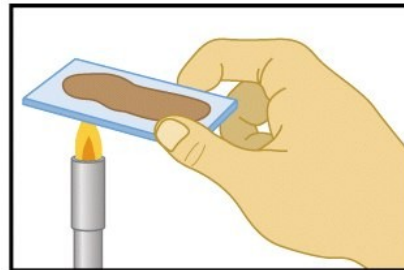
Simple Stain

Spread culture in thin film over slide

I. Preparing a smear

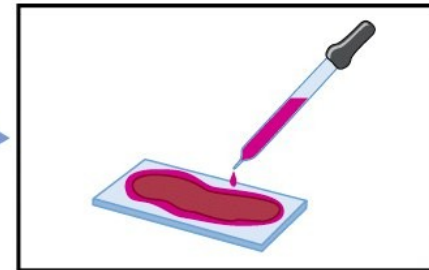


Dry in air



Pass slide through flame to fix

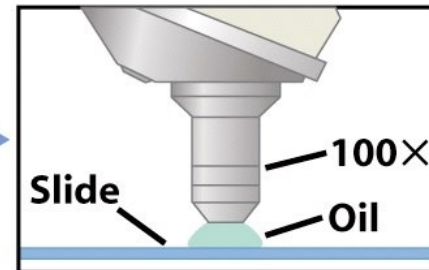
II. Heat fixing and staining



Flood slide with stain; rinse and dry



III. Microscopy

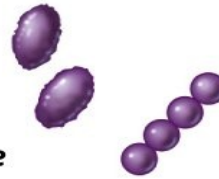


Place drop of oil on slide; examine with 100X objective

Differential **The Gram Stain**

Step 1

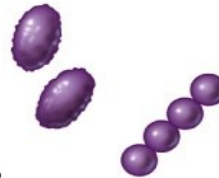
Result:
All cells purple



Flood the heat-fixed smear with crystal violet for 1 min

Step 2

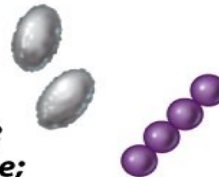
Result:
All cells remain purple



Add iodine solution for 1 min

Step 3

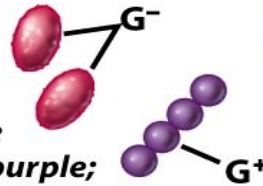
Result:
*Gram-positive cells are purple;
gram-negative cells are colorless*



Decolorize with alcohol briefly — about 20 sec

Step 4

Result:
*Gram-positive (G^+) cells are purple;
gram-negative (G^-) cells are pink to red*

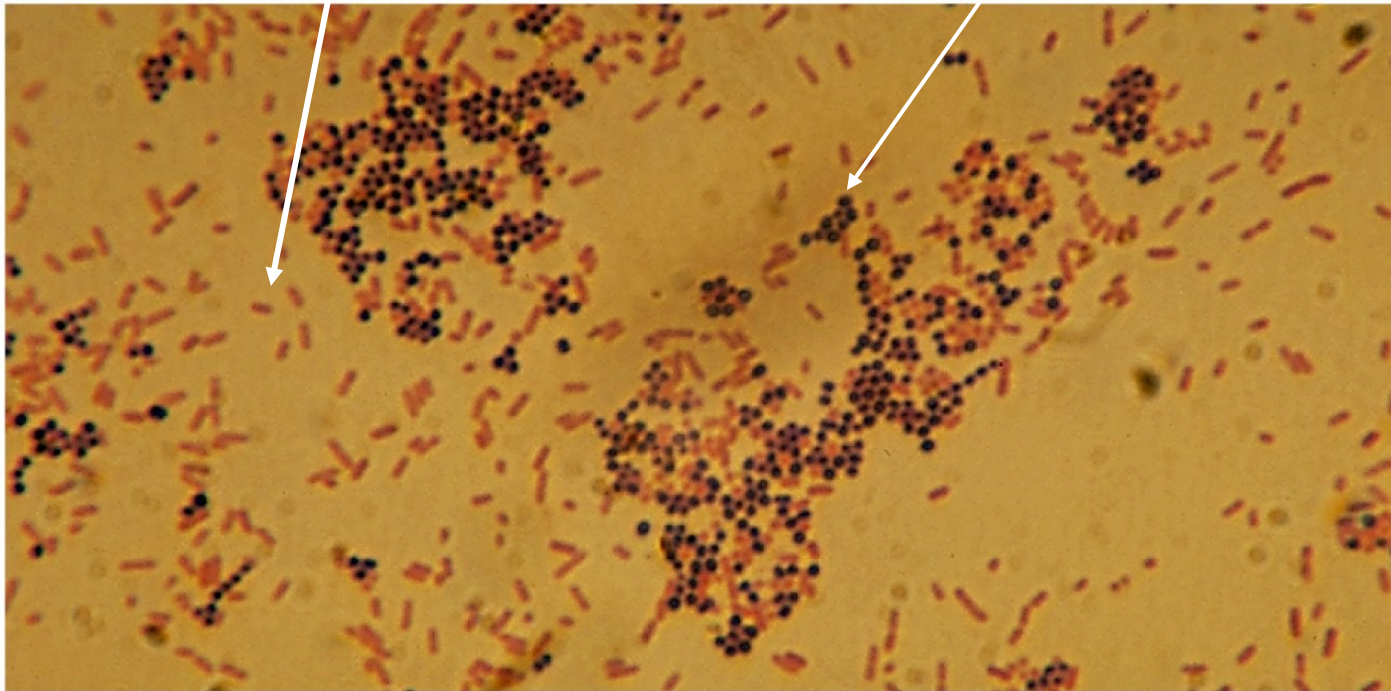


Counterstain with safranin for 1–2 min

Example for Gram Stain

E. coli (g-)

S. aureus (g+)



Leon J. Lebeau

Figure 4-4b Brock Biology of Microorganisms 11/e
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Differential Stain

A differential stain consists of two or more dyes and is used in the procedure to identify bacteria. Two of the most commonly used differential stains are the Gram stain and the Zehl-Nelson acid-fast stain.

In 1884 Hans Christian Gram, a Danish physician, developed the Gram stain. Gram-stain is a method for the differential staining of bacteria. Gram-positive microorganisms stain purple. Gram-negative microorganisms stain pink.

Staphylococcus aureus, a common bacterium that causes food poisoning, is gram-positive, *Escherichia coli* is gram-negative.

The *Ziehl-Neelsen* acid-fast stain, developed by Franz Ziehl and *Friedrick Neelsen*, a red dye that attracts to the waxy material in the cell walls of bacteria such as *Mycobacterium tuberculosis*, which is the bacterium that causes tuberculosis, and *Mycobacterium leprae*, which is the bacterium that causes leprosy.

Microorganisms that retain the Ziehl-Neelsen acid-fast stain are called acid-fast. Those that do not retain it turn blue because the microorganism doesn't absorb the Ziehl-Neelsen acid-fast stain.

Special Stains:

Use color to isolate and identify specific parts of a bacterium

Special stains are paired to dye specific structures of microorganisms such as endospores, flagella and gelatinous capsules. One stain in the pair is used as a negative stain. A negative stain is used to stain the background of the microorganism.

Acid-fast staining

Particularly useful for staining members of the genus *Mycobacterium*

e.g., *Mycobacterium tuberculosis* – causes tuberculosis

e.g., *Mycobacterium leprae* – causes leprosy

high lipid content in cell walls is responsible for their staining characteristics

Staining Specific Structures

Spore staining

double staining technique

bacterial endospore is one color and vegetative cell is a different color



Flagella staining

mordant applied to increase thickness of flagella

How to Gram-stain a specimen

Observing Microorganisms

1. Prepare the specimen using the heat fixation process
2. Place a drop of crystal violet stain on the specimen.
3. Apply iodine on the specimen using an eye dropper. The iodine helps the crystal violet stain adhere to the specimen. Iodine is a mordant which is a chemical that fixes the stain to the specimen.
4. Wash the specimen with ethanol or alcohol-acetone solution, then wash with water.
5. Wash the specimen to remove excess iodine. The specimen appears purple in colour.
6. Apply the safranin stain to the specimen using an eyedropper.
7. Wash the specimen.
8. Use a paper towel and blot the specimen until the specimen is dry.
9. the specimen is ready to be viewed under the microscope. Gram-positive bacteria appear purple and gram-negative bacteria appear pink.

How to apply the Ziehl-Neelsen acid-fast stain to a specimen.

1. Prepare the specimen.
2. Apply the red dye carbon-fuchsin stain generously using an eyedropper.
3. Let the specimen sit for a few minutes.
4. Warm the specimen over steaming water. The heat will cause the stain to penetrate the cell wall.
5. Wash the specimen with an alcohol-acid or acid-alcohol decolorizing solution consisting of 3 percent hydrochloric acid and 93 percent ethanol. The hydrochloric acid will remove the color from non-acid-fast cells and the background. Acid-fast cells will stay red because the acid cannot penetrate the cell wall.
6. Apply methylene blue stain on the specimen using an eyedropper.

The Prokaryotic Cell Wall

Why study bacterial cell walls?

They are essential structures in bacteria.

They are made of chemical components found nowhere else in nature.

They may cause symptoms of disease in animals.

They are the site of action of some of our most important antibiotics.

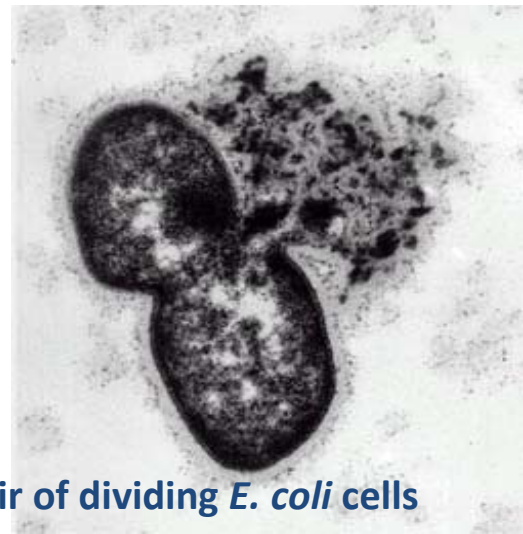
Primary function of the bacterial cell wall

To prevent rupture or osmotic lysis of the cell protoplast

Shape provided

Contribute to pathogenicity (e.g. LPS)

Protection from toxic compounds



Lysis of a pair of dividing *E. coli* cells

Chemical nature of bacterial cell walls

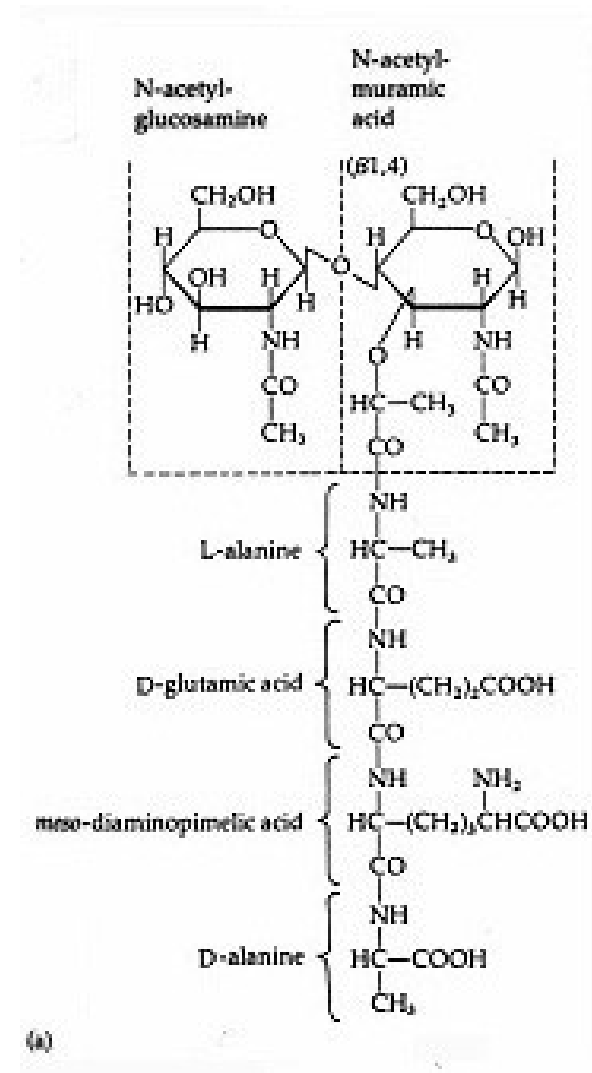
Gram-positive cell wall is thick homogeneous monolayer

Gram-negative cell wall is thin heterogeneous multilayer

Bacterial cell walls always contain murein, which is a type of peptidoglycan

Chemical nature of murein accounts for the function of the cell wall

Murein is only found in the cell walls of bacteria



E. coli peptidoglycan

Chemical nature of bacterial cell walls

Peptidoglycan is made up of 2 amino sugars

N-acetyl-glucosamine = **G**

N- acetylmuramic acid = **M**

4 amino acids

L-alanine = **L-ala**

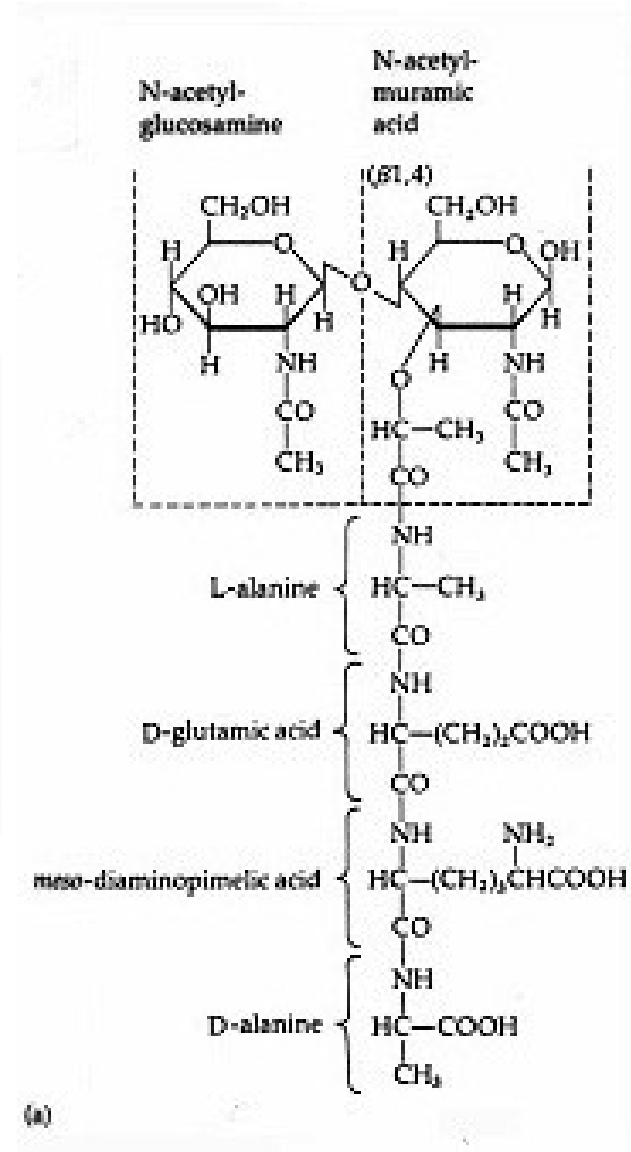
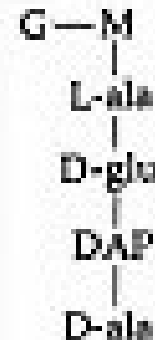
D-glutamic acid = **D-glu**

diaminopimelic acid = **DAP**

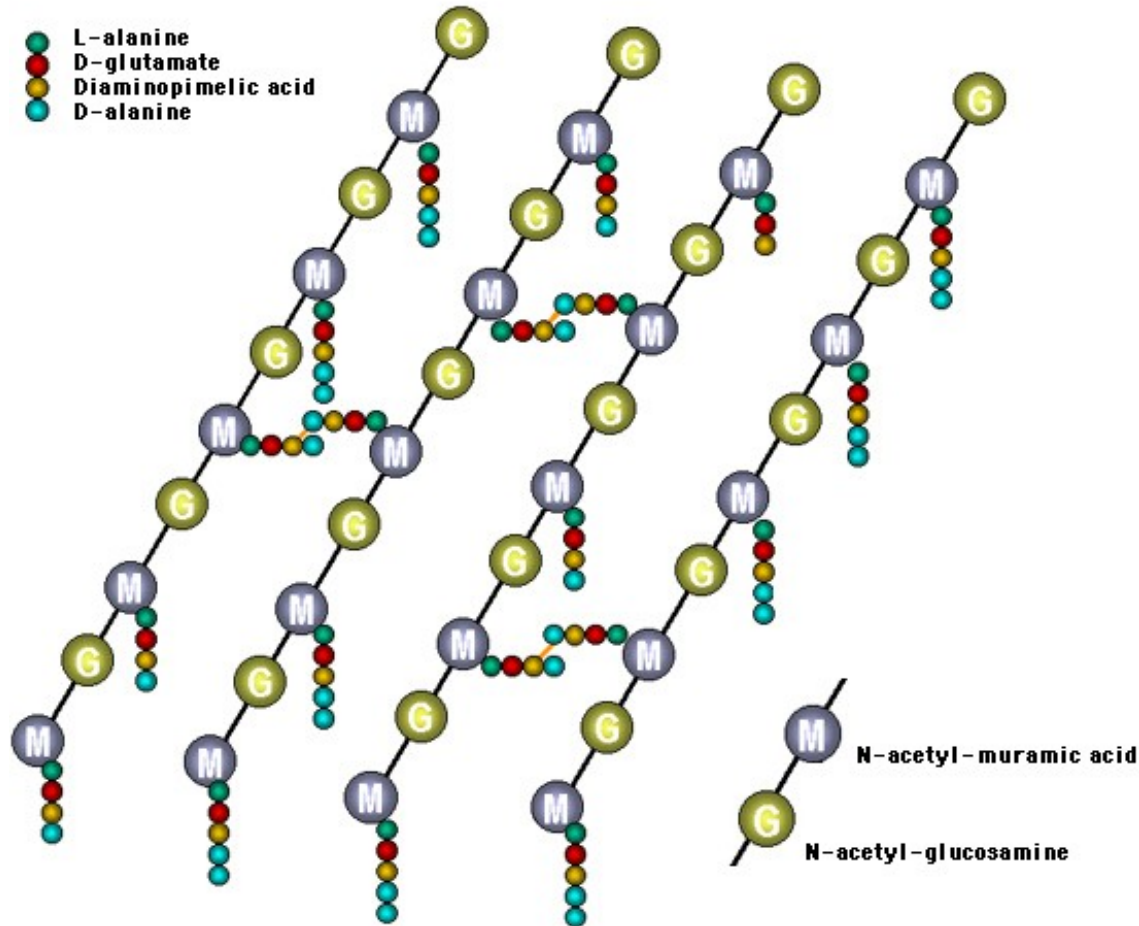
D-alanine = **D-ala**

(D-amino acids and meso-diaminopimelic acid not found in proteins)

Alternating D- and L- amino acids

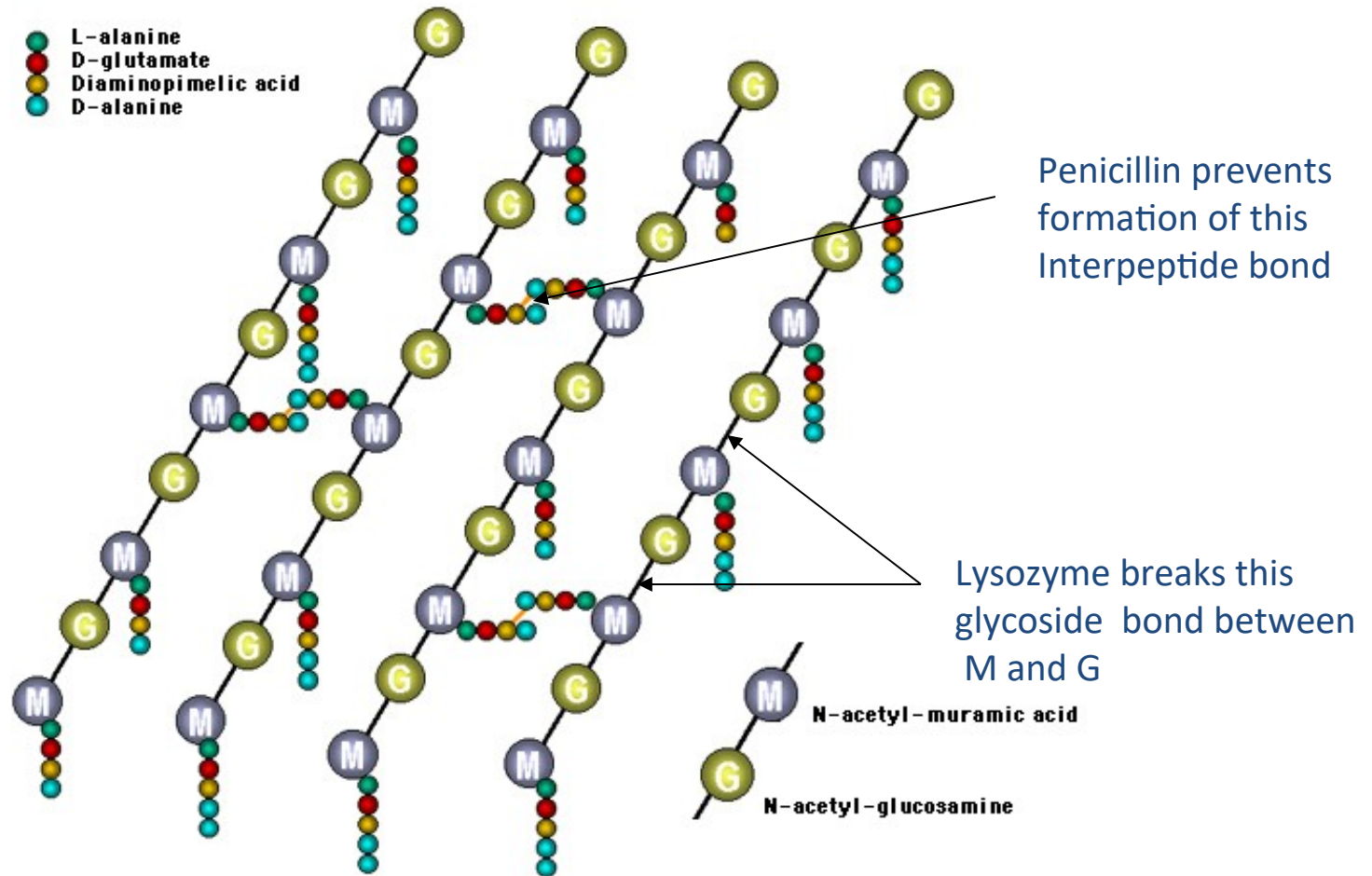


Chemical nature of bacterial cell walls



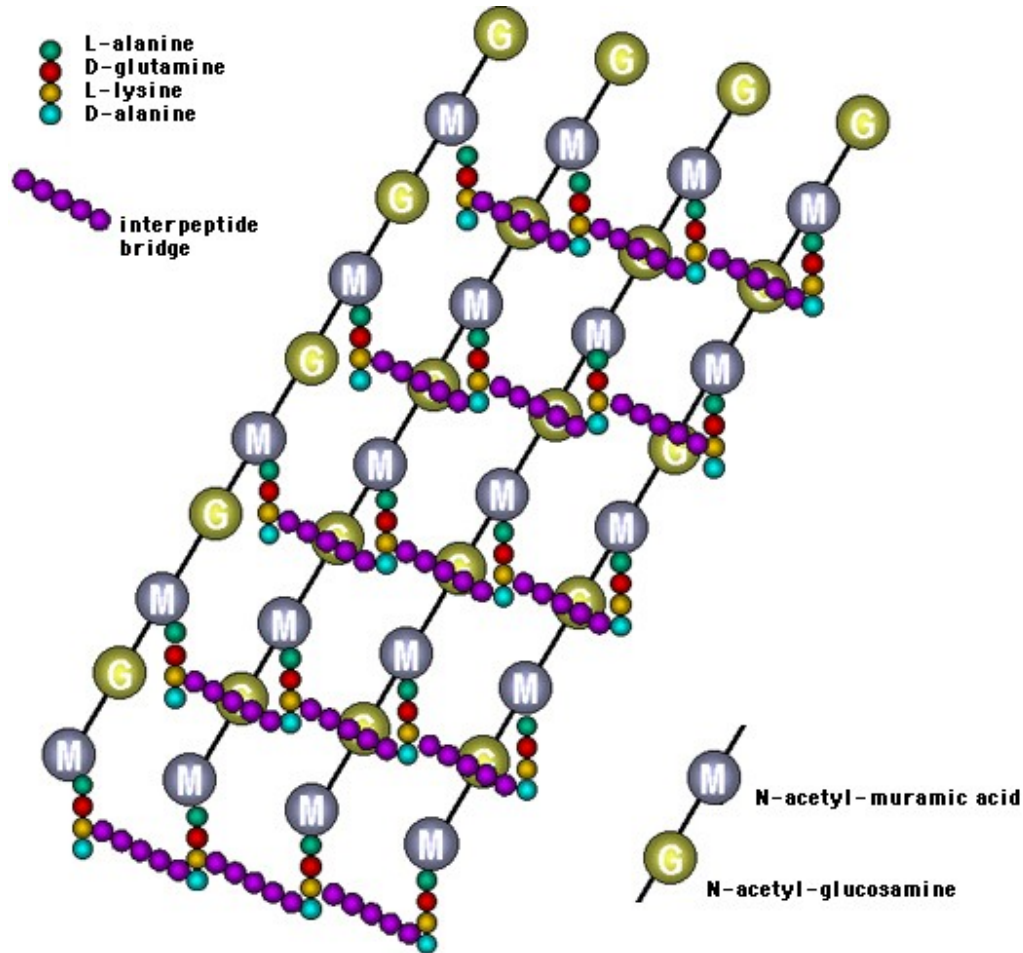
Gram-negative murein. Murein is a polymer of the peptidoglycan subunit. The sugars form the glycan backbone (G-M-G-M-etc.) and the amino acids comprise the peptide side chains of the molecule.

Chemical nature of bacterial cell walls



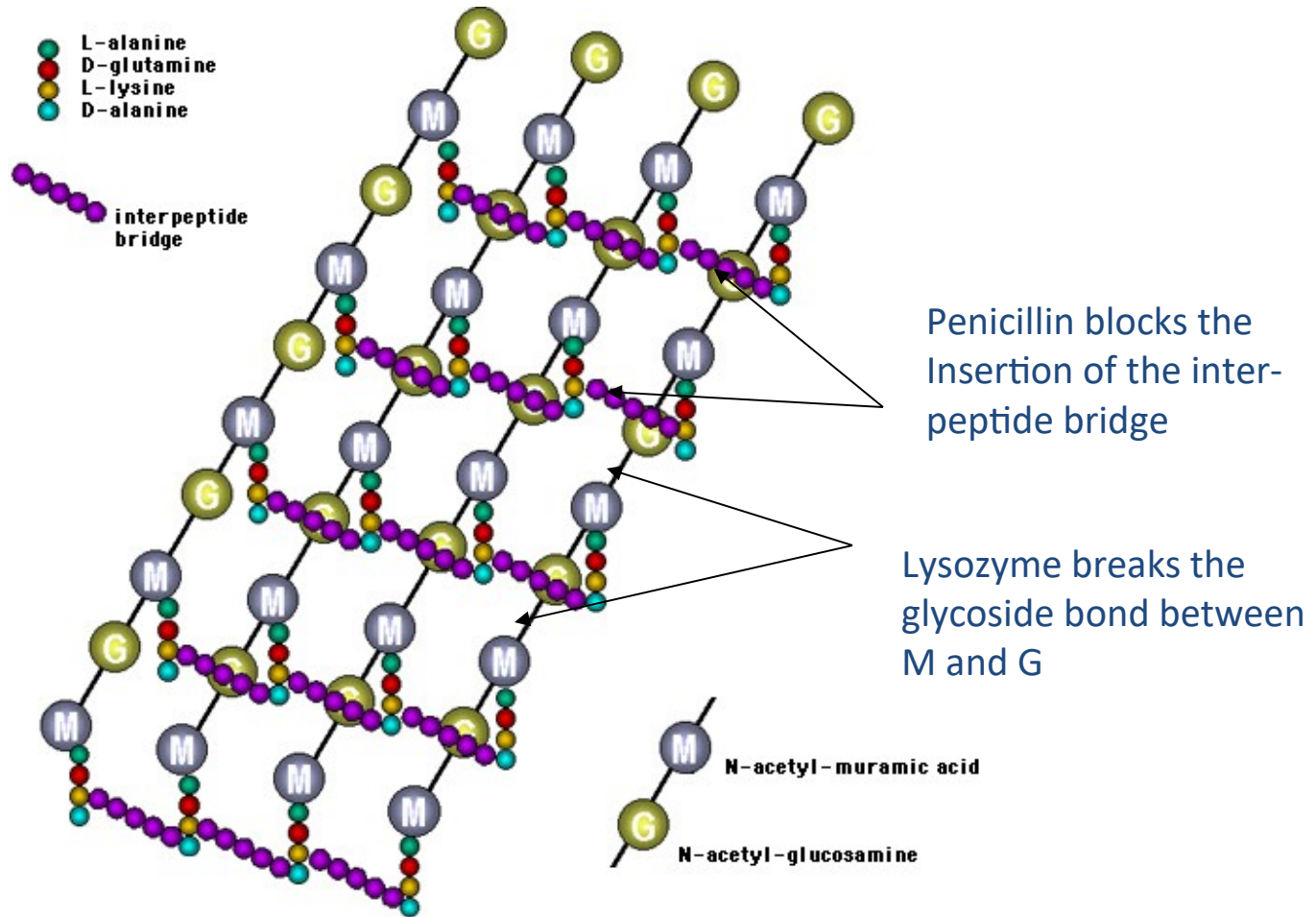
Gram-negative murein showing the sites of action of the antibiotic penicillin and the enzyme lysozyme

Chemical nature of bacterial cell walls



Gram-positive murein has a thicker glycan backbone and there are interpeptide bridges that join amino acid side chains together.

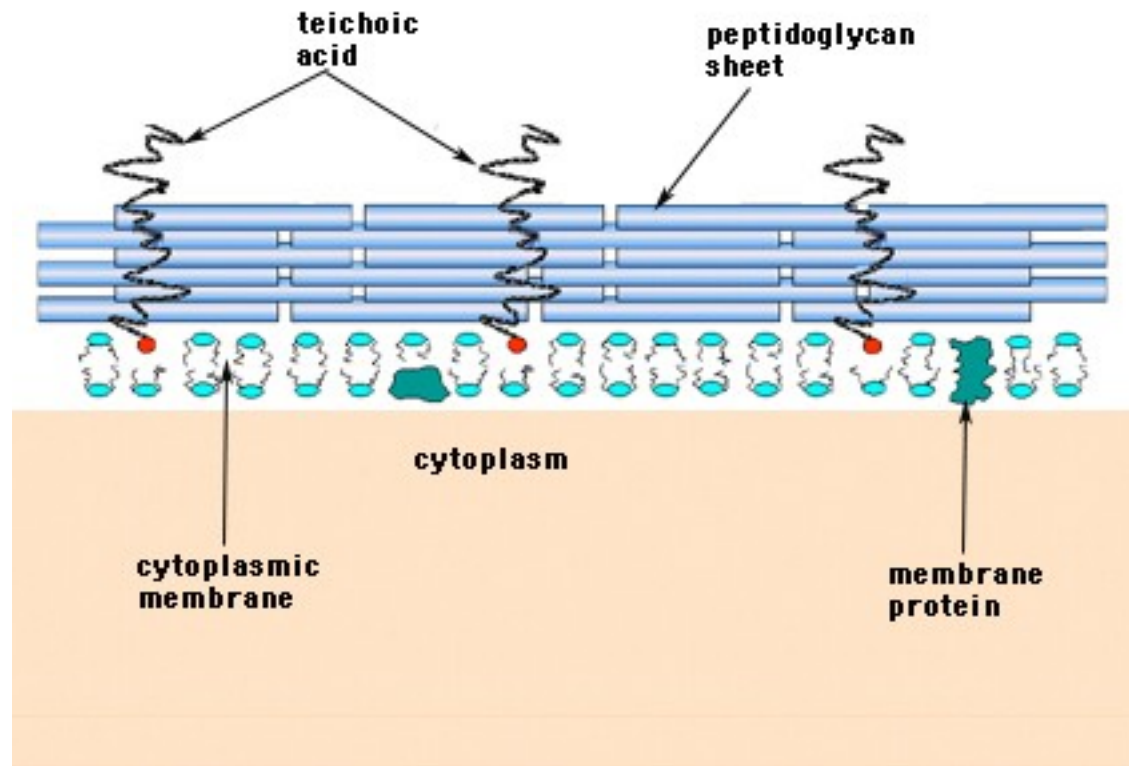
Chemical nature of bacterial cell walls



Gram-positive murein showing the sites of action of the antibiotic penicillin and the enzyme lysozyme

Other characteristics of bacterial cell walls

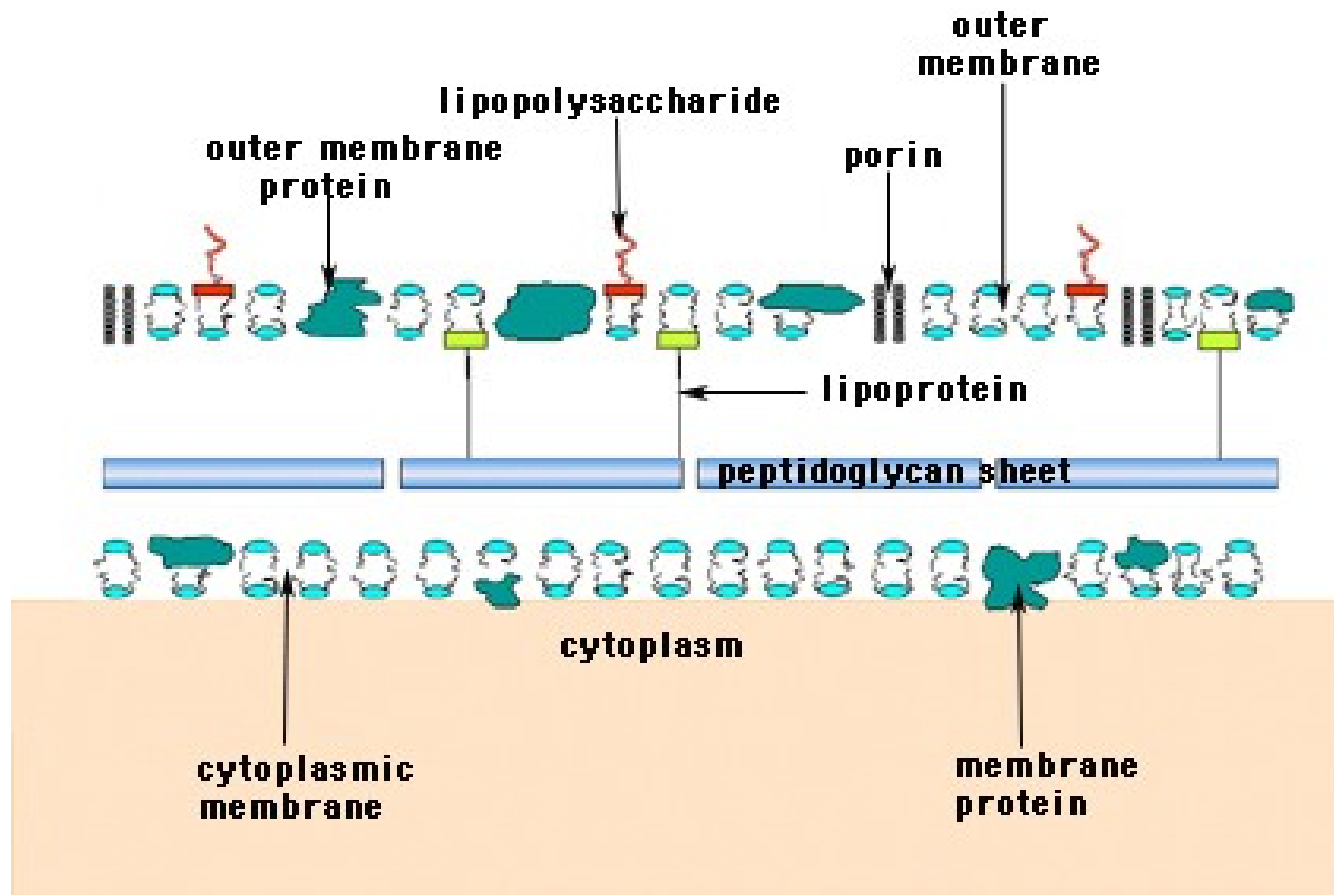
Gram-positive cell walls contain teichoic acids



Teichoic acids are thought to stabilize the Gram positive cell wall and may be used in adherence.

Other characteristics of bacterial cell walls

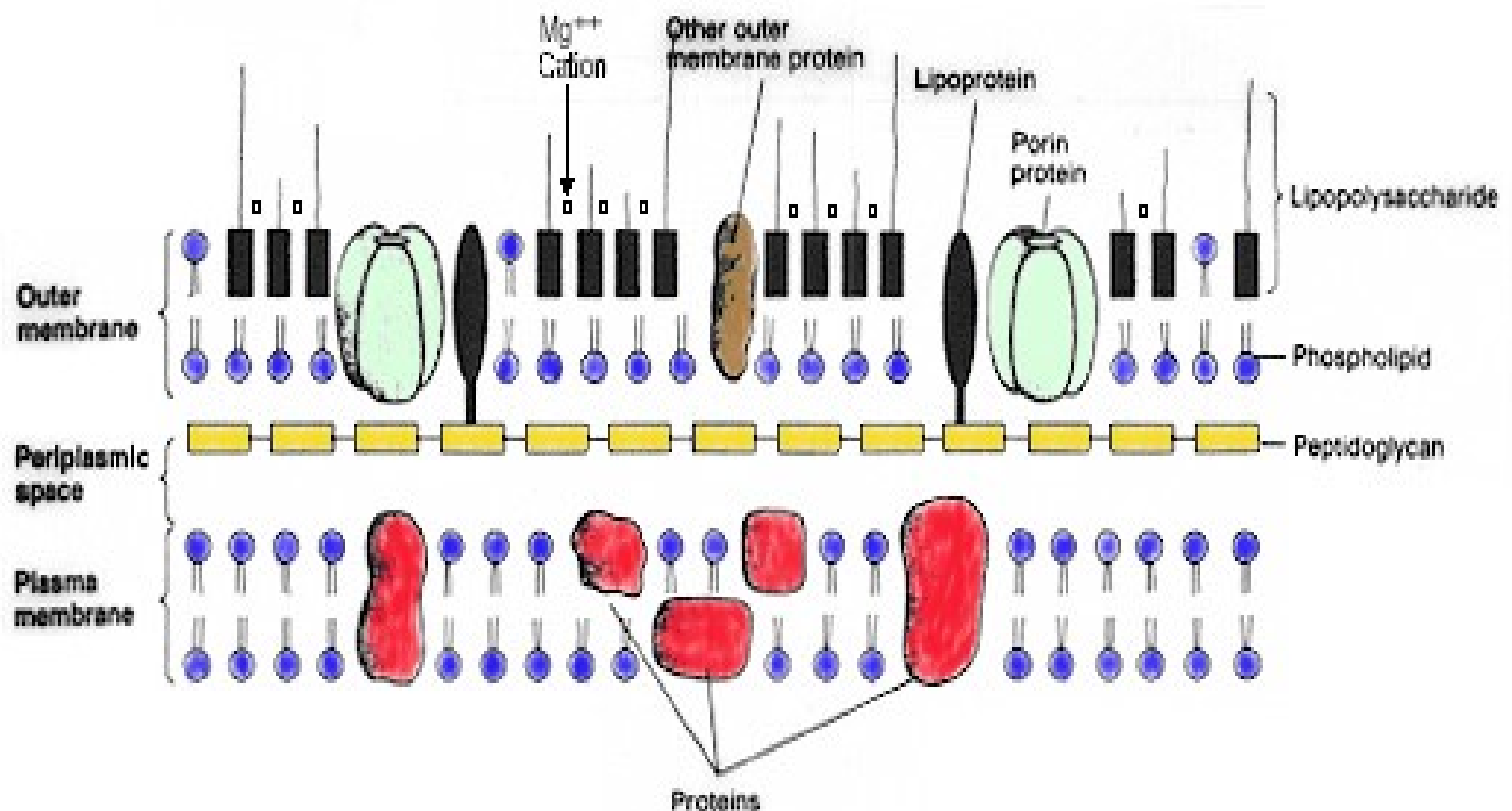
Gram-negative cell walls include an outer membrane



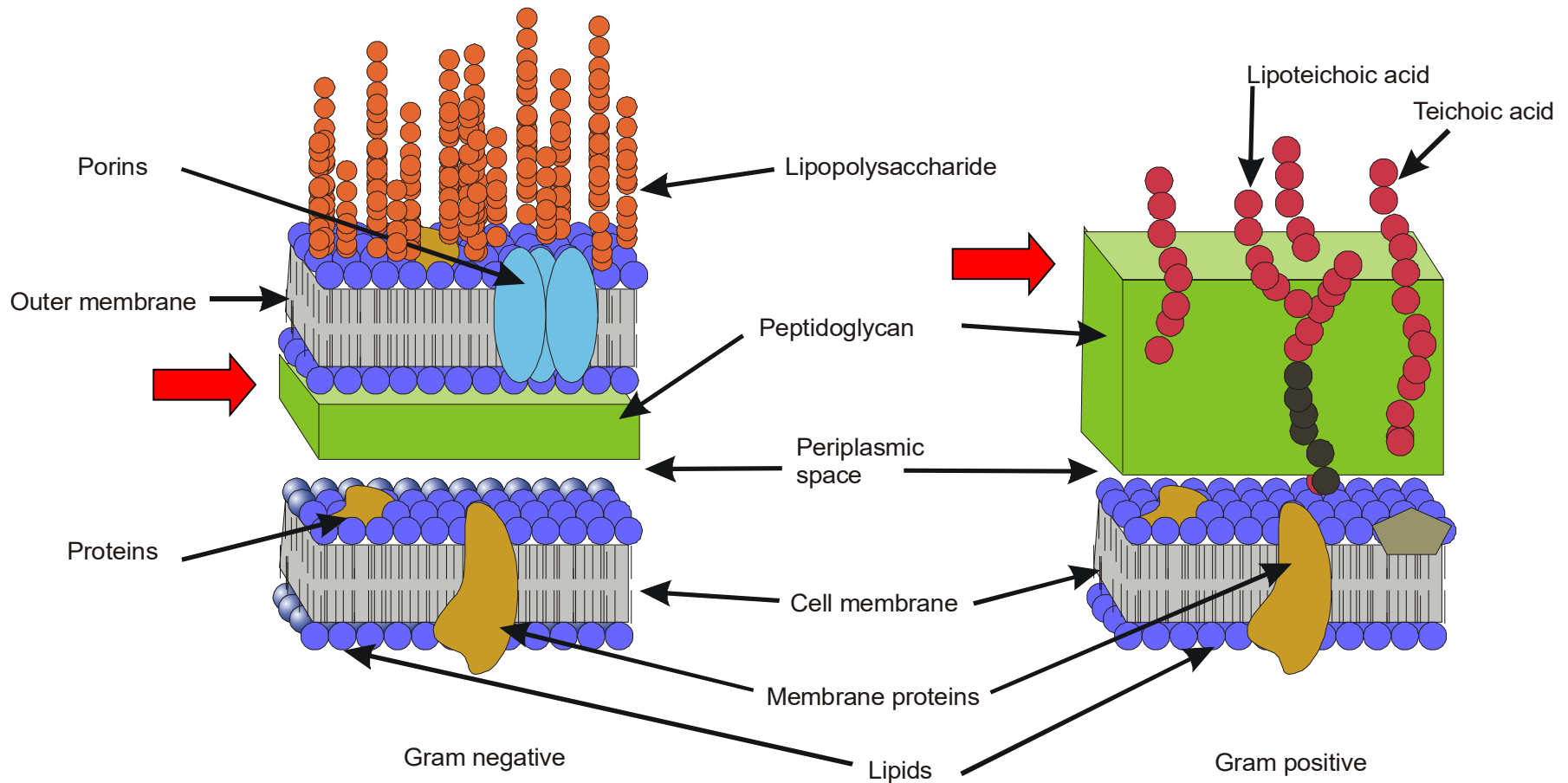
Other characteristics of bacterial cell walls

Outer membrane of Gram-negatives has two important properties It protects the cells from permeability by many substances including penicillin and lysozyme.

It is the location of lipopolysaccharide (endotoxin) which is toxic for animals.



Cell surface meets the outside world



Cell wall - The cell wall is a rigid structure composed of peptidoglycan that maintains the characteristic shape of the cell.

- permeable to small molecules (<15,000)

Table: Correlation of the Gram stain with properties of bacterial cell walls

Property	Gram-positive	Gram-negative
Thickness of wall	thick (20-80 nm)	thin (10 nm)
Number of layers	1	2-3
Peptidoglycan (murein) content	>50%	10-20%
Teichoic acids in wall	present	absent
Protein/lipoprotein content	0-3%	>50%
Lipopolysaccharide content	0	13
Sensitivity to penicillin	sensitive	resistant
Sensitivity to lysozyme	sensitive	resistant

Differences between Gram-positive and Gram-negative Bacteria

Gram-positive cell wall

- Composed of THICK PTG (murein layer)

- Single layer – 20-80 nm

- Located outside of PM

- Contains large amounts of teichoic acid

Gram-negative cell wall

- THIN PTG

- 1-3nm

- Surrounded by an outer-membrane located outside of PM

- Contains lipopolysaccharide (LPS = endotoxin)

Periplasmic Space

A gap between outer-membrane and plasma membrane of Gram-negative bacteria

A gap between cell wall and plasma membrane of Gram-positive bacteria

Filled with periplasm

Proteins used for metabolism, protection and export

Gram-Positive Strategy

Polymeric chains of PTG stabilized by cross-connecting bridges of amino acids in Gram-positive bacteria

- Cross-links form an interbridge

- Pentaglycine peptide bridge utilized

- Covalently link D-ala (4th amino acid) of tetrapeptide attached to NAM with L-lysine (3rd amino acid) of another tetrapeptide

- Strengthens cell wall – Dense network – THICK

Often contain **teichoic acids**

- Linked to muramic acid of PTG via phosphodiester bonds

- Two major forms

 - Ribitol teichoic acid**

 - Long polymer of ribitol

 - 5 carbon sugar alcohol

 - Glycerol teichoic acid**

 - Long polymer of glycerol

 - 3 carbon sugar alcohol

- Sometimes linked to PM lipids (lipoteichoic acids)

- Extend through peptidoglycan layer to external surface

- Highly antigenic

 - Antibodies often used for serological identification of different strains of Gram-positive bacteria

Gram-Negative Strategy

Carboxyl group of D-ala (#4) is covalently linked too amino group of diaminopimelic acid (#3)

No pentaglycine bridge is involved

tetrapeptides themselves are the bridge

Often have “free” tetrapeptides

Lipoproteins sometimes bound to NAM

Thinner than Gram-positive PTG components

More complex chemically than Gram-positive cell walls

Less PTG (only 5-10%, much thinner – 1-3nm, usually a monolayer)

Second membrane structure located exterior to PTG = Outer-membrane (OM)

Composed of protein, phospholipid, lipoprotein and lipopolysaccharide (LPS)

Braun's lipoprotein – anchors OM to PTG

LPS = toxic to animals

LPS - Three Major Components

Lipid A

- Two glucosamine sugar derivatives linked to fatty acids

- Phosphate or pyrophosphate attached

- Responsible for toxicity

- Buried in OM

Core polysaccharide

- Linked to Lipid A

- Often contains unusual sugars (e.g. heptuose)

- Contains phosphate

O antigen

- somatic antigen

- Short polysaccharide often with unusual sugars (e.g. mannose, galactose)

- Different types depending on strain

Functions of LPS

Stabilization of OM

Provide net negative charge on the cell

Function of OM

Protective barrier

Inhibits or regulates toxic compounds, metabolites

Permits entry of small molecules through porin proteins

Prevents loss of periplasmic enzymes

How the Gram Stain Works

Gram-positive and Gram-negative bacteria stain differently because of the composition of their cell walls

PTG layer retains crystal violet more efficiently in Gram-positive bacteria because of its thickness and degree of cross-linking (smaller pores)

Crystal violet/iodine complex more easily removed from Gram-negative bacteria

What makes Archaeobacteria unique?

Archaeobacteria (unlike eubacteria) do NOT contain PTG

Possess unusual lipids in their membranes

**Branched chain hydrocarbons connected to glycerol
by ether links**

May also contain

Pseudomurein

Polysaccharides

Glycoproteins

Differences in tRNA and RNA polymerases structure

Diverged evolutionarily from eubacteria before the origin of eukaryotes

How the Gram Stain Works

Mycoplasma lack a cell wall

They use sterols for strength

	Microscopic Appearance of Cell		Chemical Reaction in Cell Wall (very magnified view)	
Step	Gram (+)	Gram (-)	Gram (+)	Gram (-)
1. Crystal violet			 Both cell walls affix the dye	
2. Gram's iodine			 Dye crystals trapped in wall	 No effect of iodine
3. Alcohol			 Crystals remain in cell wall	 Cell wall partially dissolved, loses dye
4. Safranin (red dye)			 Red dye has no effect	 Red dye stains the colorless cell